



Machine Learning

Maximum Likelihood Estimation(MLE)

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https://github.com/safayani/machine_learning_course

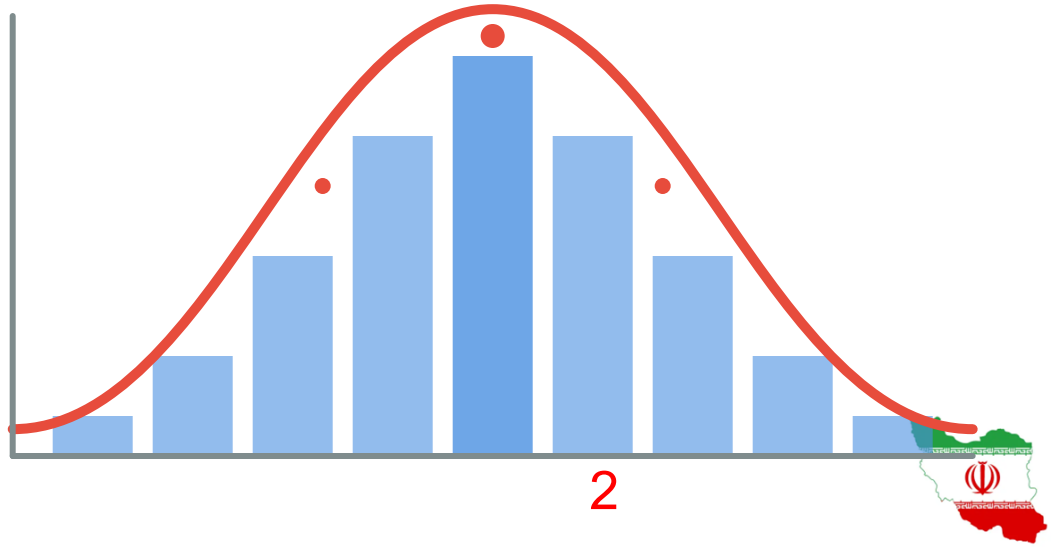
https://git.iut.ac.ir/safayani/courses/-/tree/main/Machine_Learning_course



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What is Data Distribution?

- **Definition:** How data points are spread across a dataset.
- **What it shows:** The frequency or probability of specific values.
- **Why it matters:** Essential for Statistics, Data Analysis, and Machine Learning.
- **The Goal:** Reveals underlying structures, patterns, and behaviors in the data.



Why data distribution is important in machine learning?

- Data generation: GAN, LLM, Chat-gpt
- Formally, an LLM learns:

$$P(x_t, \mid, x_1, x_2 \ldots x_{t-1})$$

  where each x_i is a token.
- Anomaly detection
- Model assumptions:
  - Many machine learning algorithms make assumptions about the distribution of the data.
  - For example:
    - Linear regression assumes that the residuals are normally distributed.
  - Violating these assumptions can lead to poor model performance.





Discrete Data vs. Continuous Data

Discrete Data

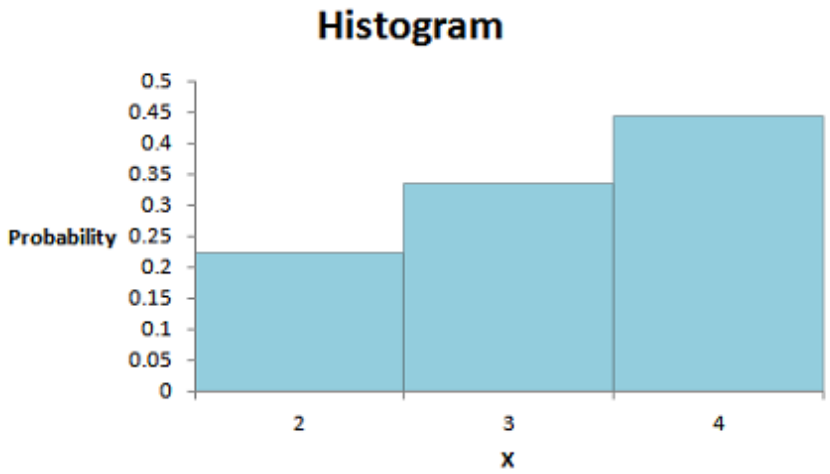
- **Definition:** Specific, finite values with no space in between.
- **Key Action:** It is **COUNTED** 🔢
- **Example:** Marital Status
  - *{Single, Married, Divorced}*
  - *(You cannot be "1.5" married)*

Continuous Data

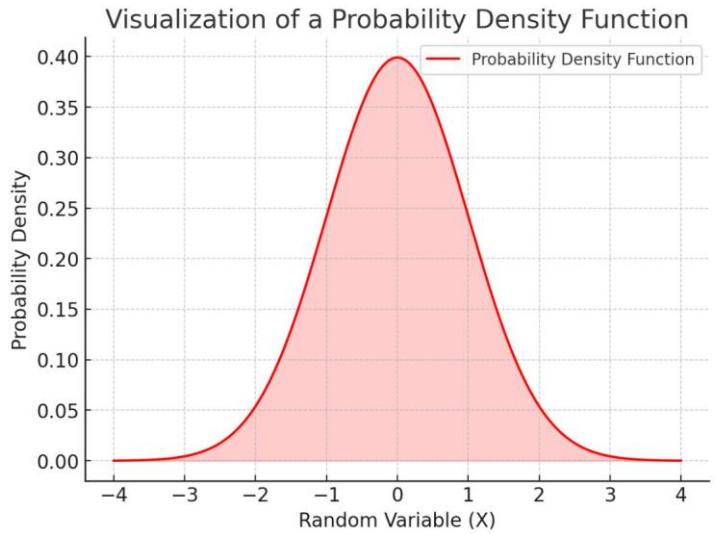
- **Definition:** Exists in a continuous space; can take *any* value.
- **Key Action:** It is **MEASURED** 📏
- **Example:** Size of a House
  - *{210.25 square meters}*
  - *(Can be broken down into infinite decimals)*

PDF vs. PMF

- **PMF**
- **Probability Mass Function**
- 📊 **Data Type:** Discrete (Countable, e.g., dice rolls)
- 🎯 **Measurement:** Probability of a *specific, exact* value.

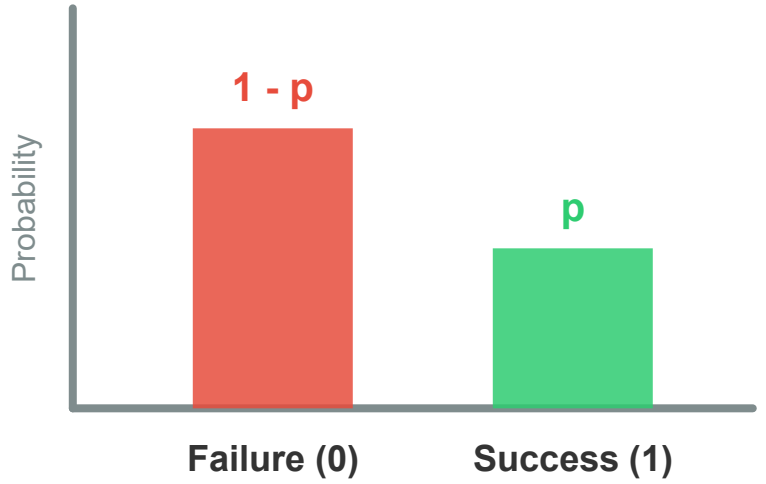


- **PDF**
- **Probability Density Function**
- 📈 **Data Type:** Continuous (Infinite possibilities, e.g., height, time)
- 🌊 **Measurement:** Probability density across a *range* of values.



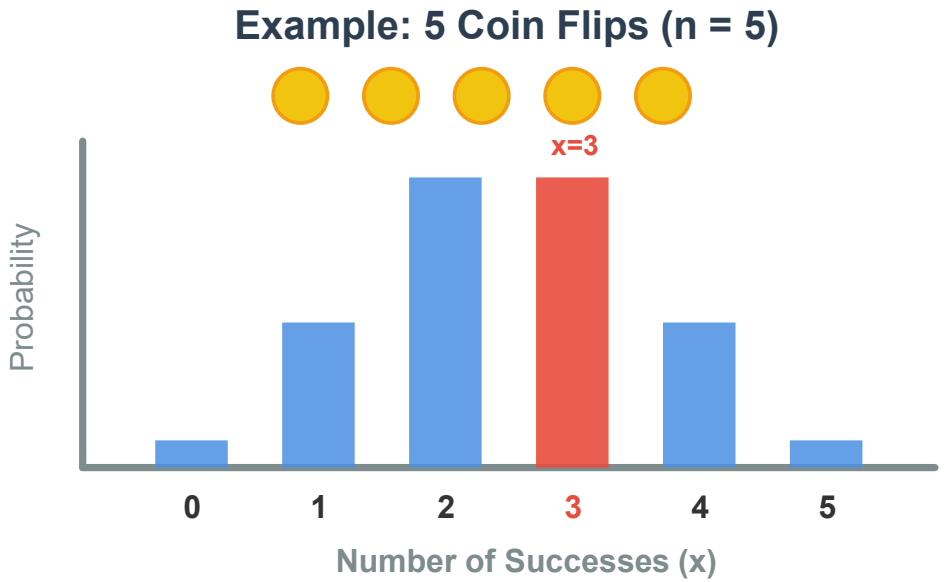
Discrete Distributions: Bernoulli Distribution

- **The Concept:** Measures the probability of exactly **one of two** outcomes (Success or Failure).
- **The Rule:** Applies to a **single trial** (one event only).
- **The Math:** A special, simplified case of the Binomial distribution where **n = 1**.
- 🪙 **Classic Example:** Flipping a coin once
  - *Heads = Success (p)*
  - *Tails = Failure (1-p)*



Discrete Distributions: The Binomial Distribution

- **The Concept:** It is the Bernoulli Distribution taken **one step further**.
- **The Outcomes:** Still binary (Success or Failure).
- **The Difference:** Instead of one single event, we are now looking at **multiple trials (*n*)**.
- **The Goal:** Measuring the probability of getting a specific number of successes (**x**) within **n** total trials.
- 🪙 **Example:** Flipping a coin 5 times.
  - *Question:* What is the probability of getting exactly 2 Heads? *(n = 5, x = 2)*



$$P_x = \frac{n!}{x!\,(n-x)!} p^x (1-p)^x$$

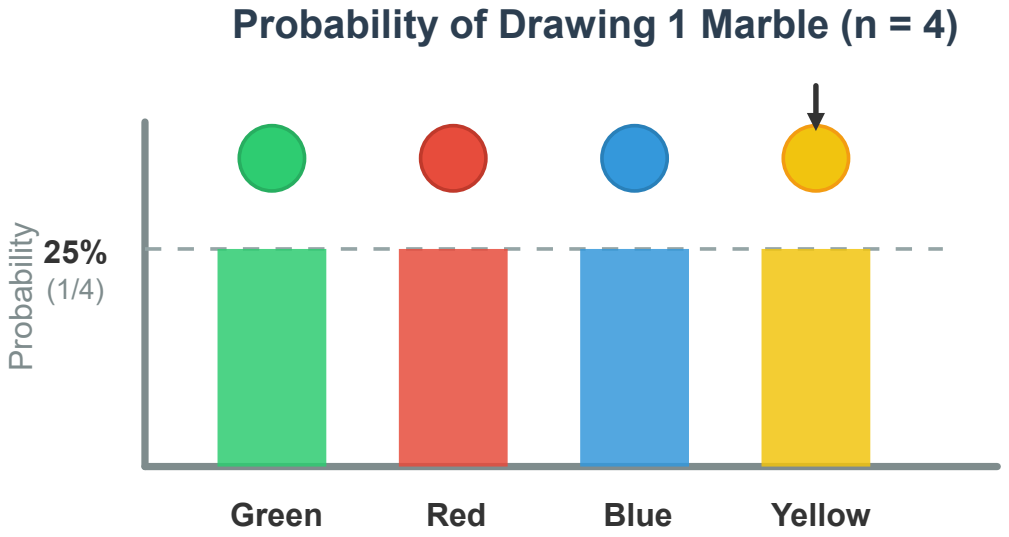
$$P_2 = \frac{5!}{2!\,(5-2)!} 0.5^2 (1-0.5)^{5-2}$$

$$P_2 = 10 * 0.25 * 0.125$$

$$P_2 = 0.3125 = 31.25\%$$

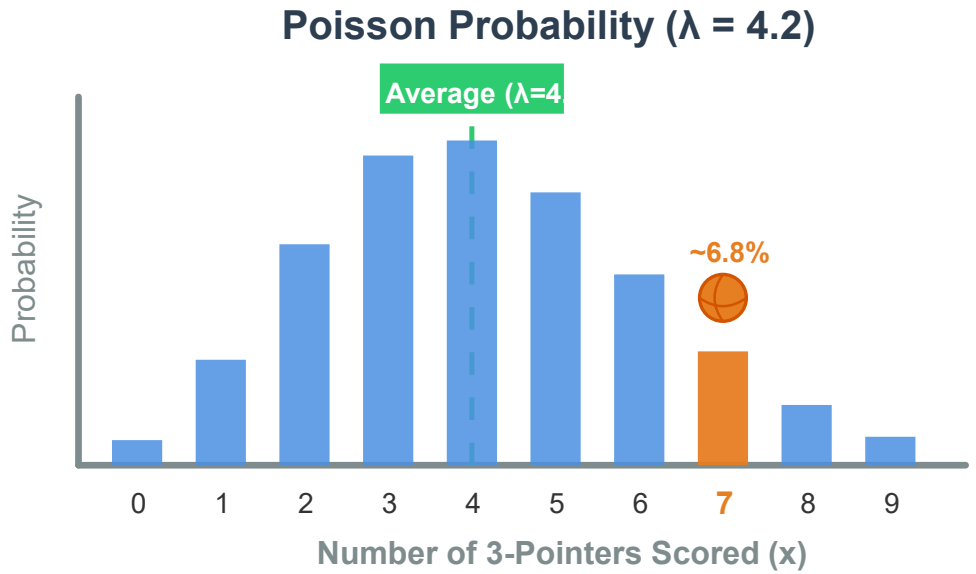
Discrete Distributions: Discrete Uniform Distribution

- **The Concept:** A simple distribution where a set of potential outcomes (n) exists.
- **The Golden Rule:** Every single outcome has an **equal likelihood** of occurring.
- **The Math:** Probability = $1/n$ (for every outcome).
- **Example:** Drawing a marble blindly from a bag of 4 marbles (Green, Red, Blue, Yellow).
  - *Question:* What are the chances of picking the Yellow marble?
  - *Answer:* Exactly **1 in 4 (25%)**.



Discrete Distributions: The Poisson Distribution

- **The Core Question:** *How many times* is an event likely to occur over a *given period of time*?
- **The Key Metric:** Defined by the **rate parameter (λ / Lambda)**.
  - *λ = The average (mean) number of occurrences in that time frame.*
- 🏀 **Example:** A basketball team's 3-point shots.
  - **The Average (λ):** They score an average of **4.2** three-pointers per quarter.
  - **The Question (x):** What is the exact probability they will score exactly **7** three-pointers in the next quarter?



Poisson PMF Formula

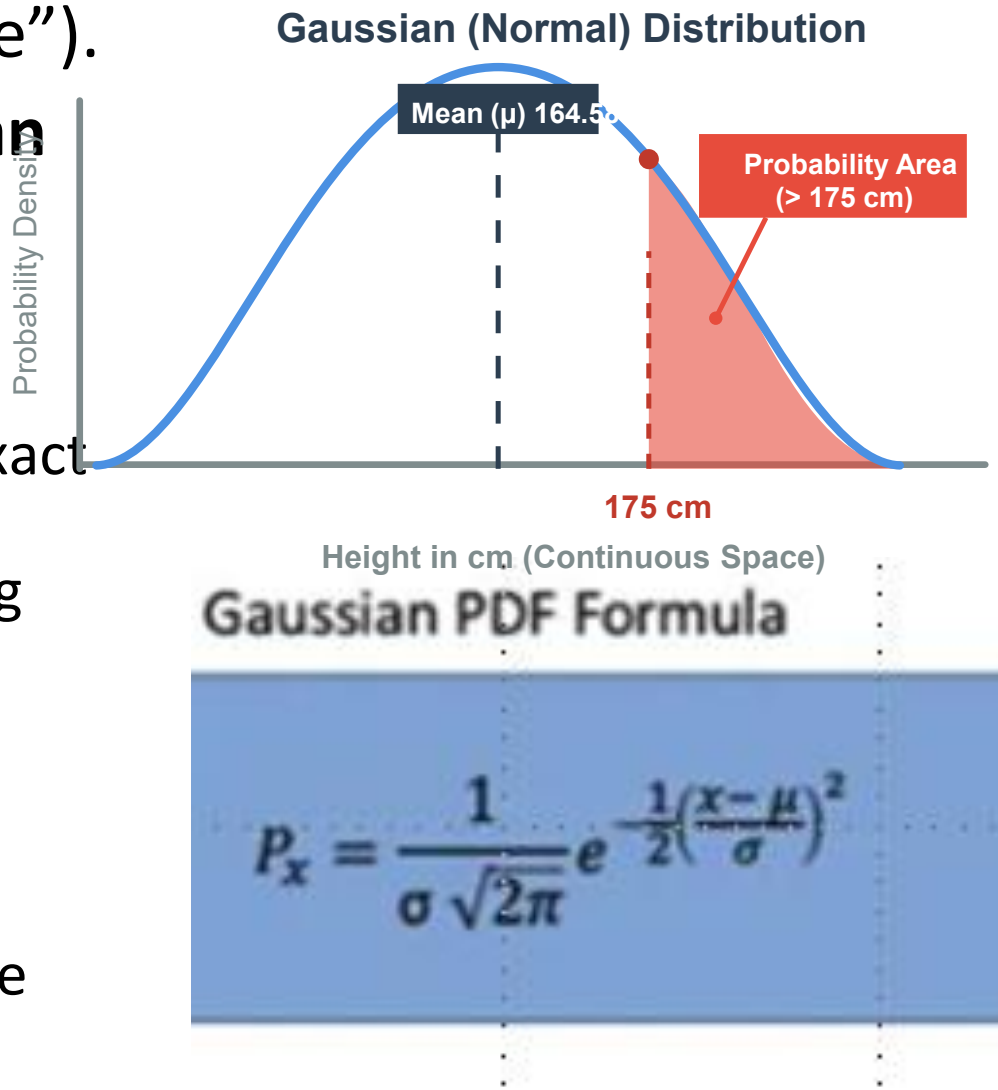
$$P_x = \frac{\lambda^x}{x!} e^{-\lambda}$$

$$P_7 = \frac{4.2^7}{7!} e^{-4.2}$$

$$P_7 = 0.0686 = 6.86\%$$

Continuos Distributions: Gaussien (Normal) Distribution

- **The Concept:** The most widely known distribution, frequently observed in nature (the “Bell Curve”).
- **The Parameters:** Defined entirely by the **Mean (μ)** and **Standard Deviation (σ)**.
- ⚠️ **The Paradigm Shift:** Because space is continuous, there are no *exact* values.
  - *Instead of asking:* “What is the likelihood of an exact value?”
  - *We ask:* “What is the likelihood of a sample falling within a **range**?”
- 📏 **Example:** Global Human Heights
  - **Mean (μ):** 164.58 cm
  - **Std Dev (σ):** 8.83 cm
  - **The Question:** What is the probability of someone being **taller than 175 cm**? *(A range!)*



The Question: What is the probability of someone being **taller than 175 cm**? *(A range!)*

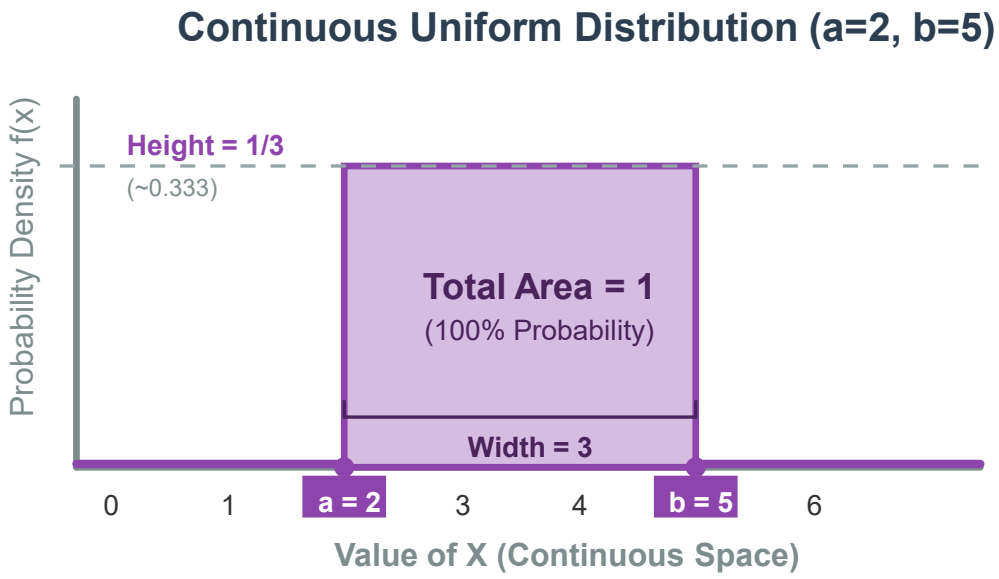
- **Step 1: Find the Z-Score (Standardization)**
  - *Formula:* $Z = (Target - Mean) / StandardDeviation$
  - *Math:* $Z = (175 - 164.58)/8.83$
  - $Z = 1.18$*) This means 175cm is 1.18 standard deviations above the average).*
- **Step 2: Calculate the Probability**
  - Using a statistical calculator or Z-Table, we look up the area to the *right* of $Z = 1.18$.
  - **The Result:** 0.1190
- 🎯 **The Final Answer:** There is an **11.9%** probability that a random person is taller than 175 cm.

STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.

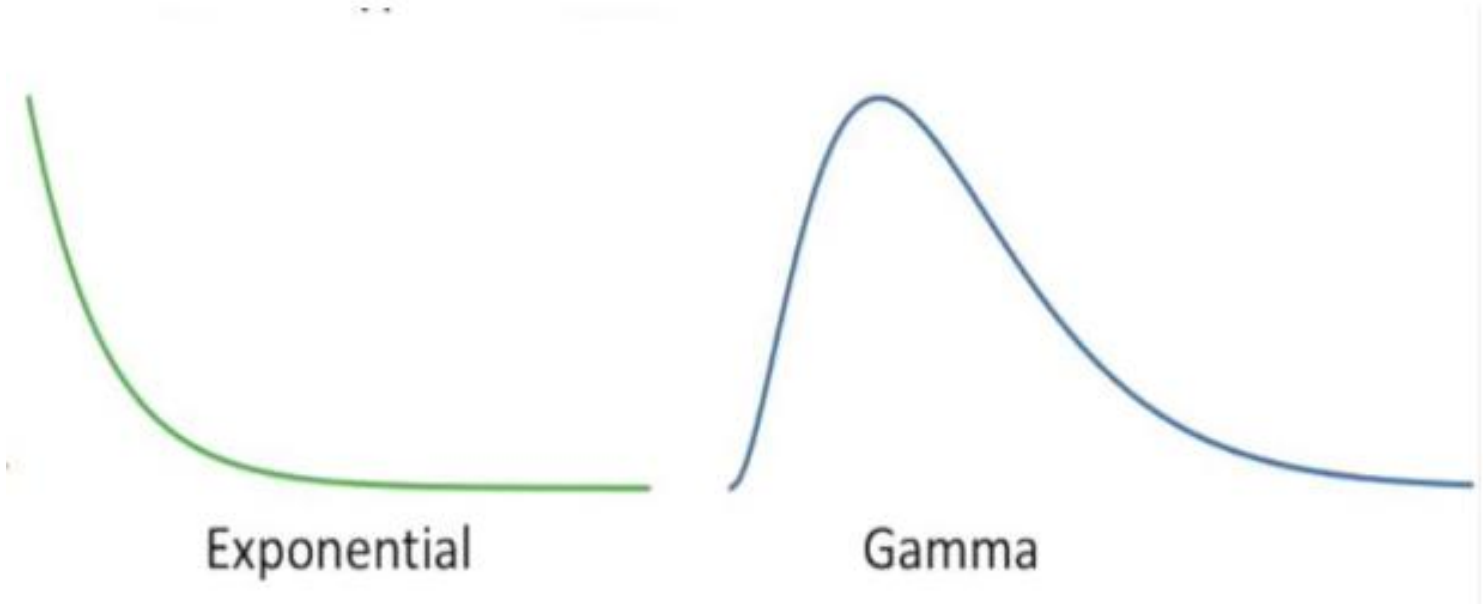
Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.50000	.50399	.50798	.51197	.51595	.51994	.52392	.52790	.53188	.53586
0.1	.53983	.54380	.54776	.55172	.55567	.55962	.56356	.56749	.57142	.57535
0.2	.57926	.58317	.58706	.59095	.59483	.59871	.60257	.60642	.61026	.61409
0.3	.61791	.62172	.62552	.62930	.63307	.63683	.64058	.64431	.64803	.65173
0.4	.65542	.65910	.66276	.66640	.67003	.67364	.67724	.68082	.68439	.68793
0.5	.69146	.69497	.69847	.70194	.70540	.70884	.71226	.71566	.71904	.72240
0.6	.72575	.72907	.73237	.73565	.73891	.74215	.74537	.74857	.75175	.75490
0.7	.75804	.76115	.76424	.76730	.77035	.77337	.77637	.77935	.78230	.78524
0.8	.78814	.79103	.79389	.79673	.79955	.80234	.80511	.80785	.81057	.81327
0.9	.81594	.81859	.82121	.82381	.82639	.82894	.83147	.83398	.83646	.83891
1.0	.84134	.84375	.84614	.84849	.85083	.85314	.85543	.85769	.85993	.86214
1.1	.86433	.86650	.86864	.87076	.87286	.87493	.87698	.87900	.88100	.88298
1.2	.88493	.88686	.88877	.89065	.89251	.89435	.89617	.89796	.89973	.90147
1.3	.90320	.90490	.90658	.90824	.90988	.91149	.91309	.91466	.91621	.91774
1.4	.91924	.92073	.92220	.92364	.92507	.92647	.92785	.92922	.93056	.93189
1.5	.93319	.93448	.93574	.93699	.93822	.93943	.94062	.94179	.94295	.94408

Continuous Distributions: Continuous Uniform Distribution

- **The Concept:** A simple distribution where every possible outcome *within a specific range* is **equally likely**.
- **The Visual:** Instead of discrete bars, the probability forms a perfect **rectangle**. The total area of this rectangle must equal 1 (100%).
- **The Math:**
  - Area = Width × Height
  - 1 = (b - a) × Height
  - Therefore, **Height = 1 / (b - a)**
- 📏 **Example:** A random variable X can take *any* value between $a = 2$ and $b = 5$.
  - **The Range (Width):** $5 - 2 = 3$
  - **The Density (Height):** $1/3$ (or ~0.333)
  - *Verification:* Base (3) × Height (1/3) = Area of 1.



Other data distributions



Statistical Methods: Maximum Likelihood Estimation (MLE)

- **The Core Question:** Given some observed data, what are the *most likely* parameters for the probability distribution that generated it?
- **The Tool: Likelihood Function**
  - This function measures how probable it is that we would observe our *exact data set* given a *specific choice* for the model parameters (e.g., a specific μ or σ).
- **The Goal of MLE:**
  - To find the one specific set of parameter values that **maximizes** the likelihood function.
  - In simple terms: MLE finds the “best-fit” parameters for your data.



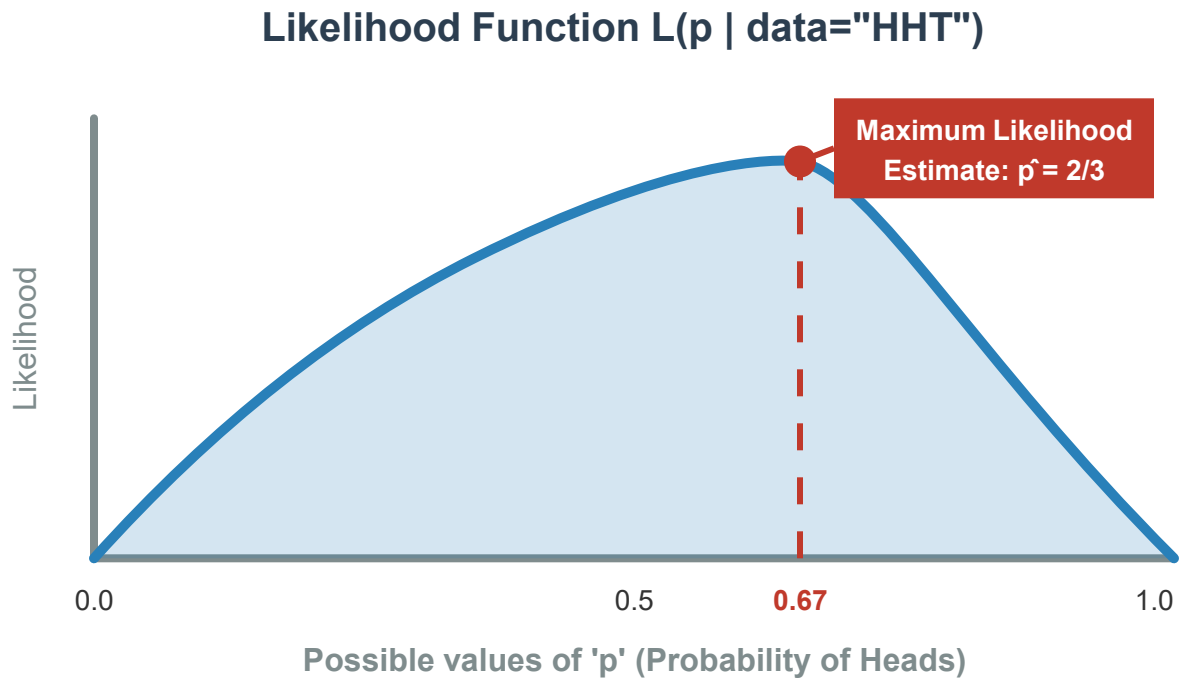


MLE: The Logic of Likelihood

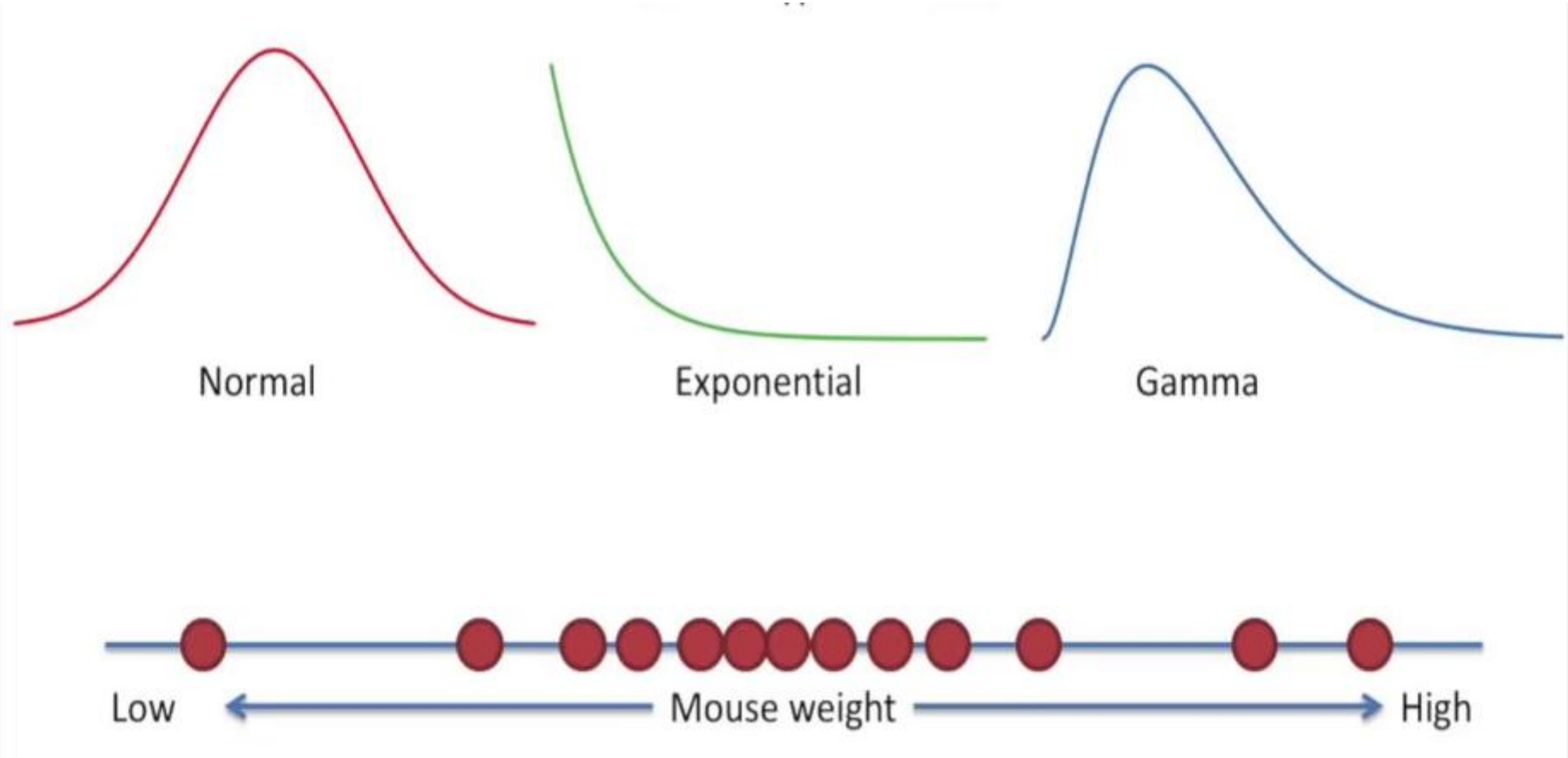
The Experiment	Testing Our Hypotheses
Our Goal: Estimate the bias (p) of an unknown coin.	We use a **Likelihood Function** to test which value of p best explains our data. A higher likelihood means the data is more probable under that hypothesis.
Our Data: We flip the coin 3 times and observe:	**Hypothesis #1: The coin is fair (p = 0.5)**
🪙 **Heads,** 🪙 **Heads,** ⚫ **Tails**	Likelihood(p=0.5) = P(H) × P(H) × P(T) = 0.5 × 0.5 × 0.5 = **0.125**
	Hypothesis #2: The coin is biased (p = 2/3)
	Likelihood(p=2/3) = P(H) × P(H) × P(T) = (2/3) × (2/3) × (1/3) ≈ **0.148**
	Conclusion: Our observed data is **more likely** to have occurred if the coin's bias was 2/3 than if it was fair. This is a better fit!

MLE: Finding the Peak Likelihood

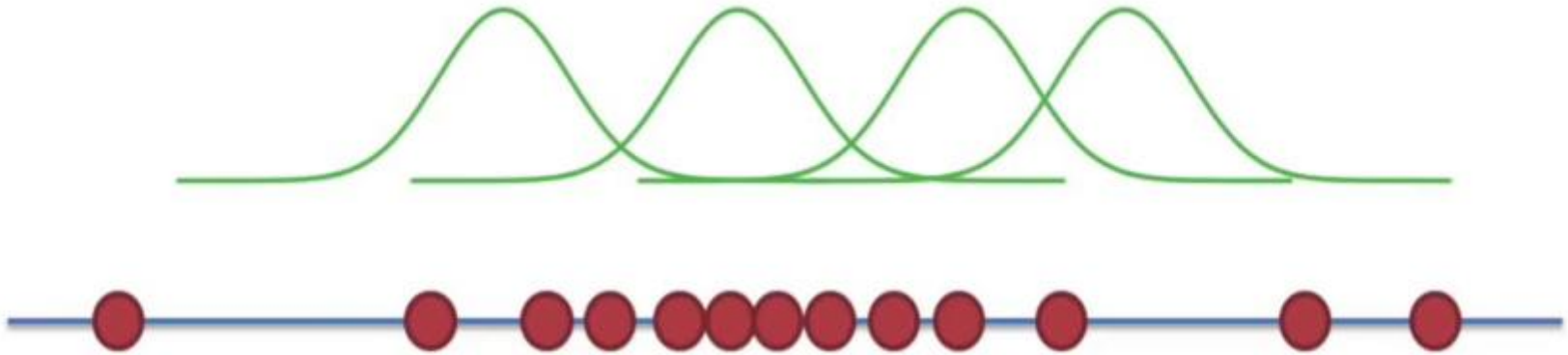
- MLE doesn't just test two options; it uses calculus to check **every possible value** of the parameter (*p*) between 0 and 1.
- It finds the *exact* value that is at the **peak** of the likelihood function.
- This peak value is called the **Maximum Likelihood Estimate (MLE)**, denoted with a "hat" (e.g., $\hat{p}$).
- **The Result:** For our data (HHT), the math shows the likelihood is maximized when:
  - **$\hat{p} = 2/3$**
- This formalizes our intuition: The most likely parameter is the one that reflects the proportions we saw in our sample data.

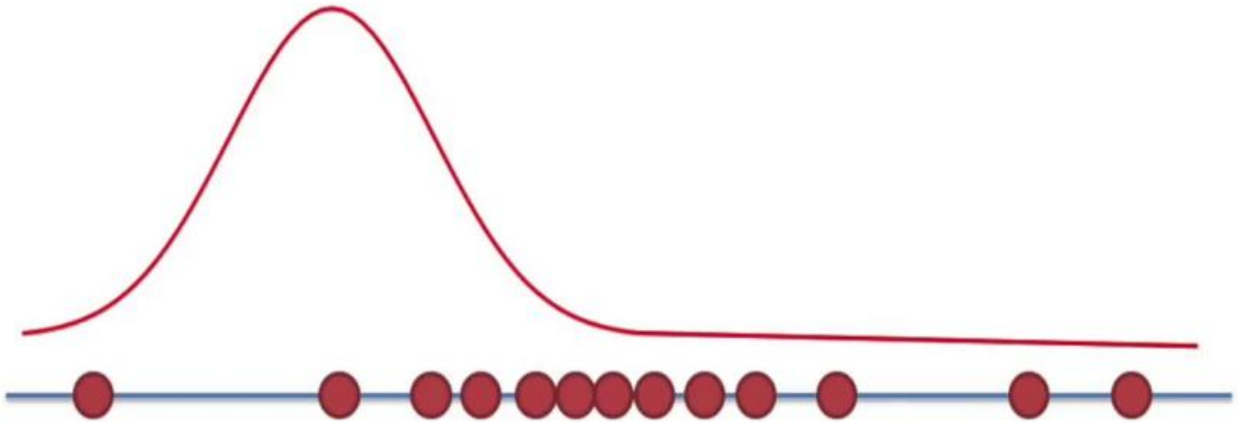


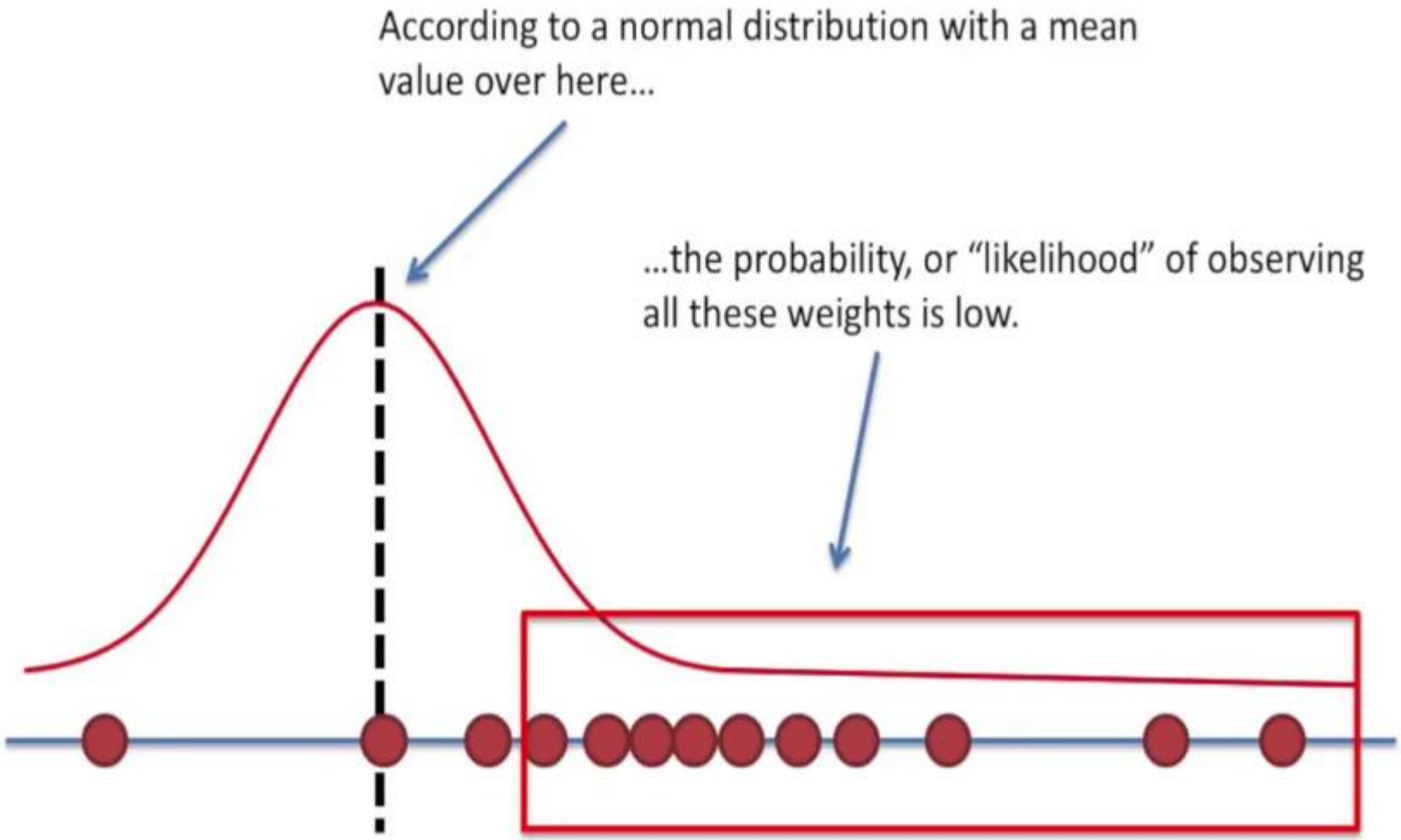
MLE example



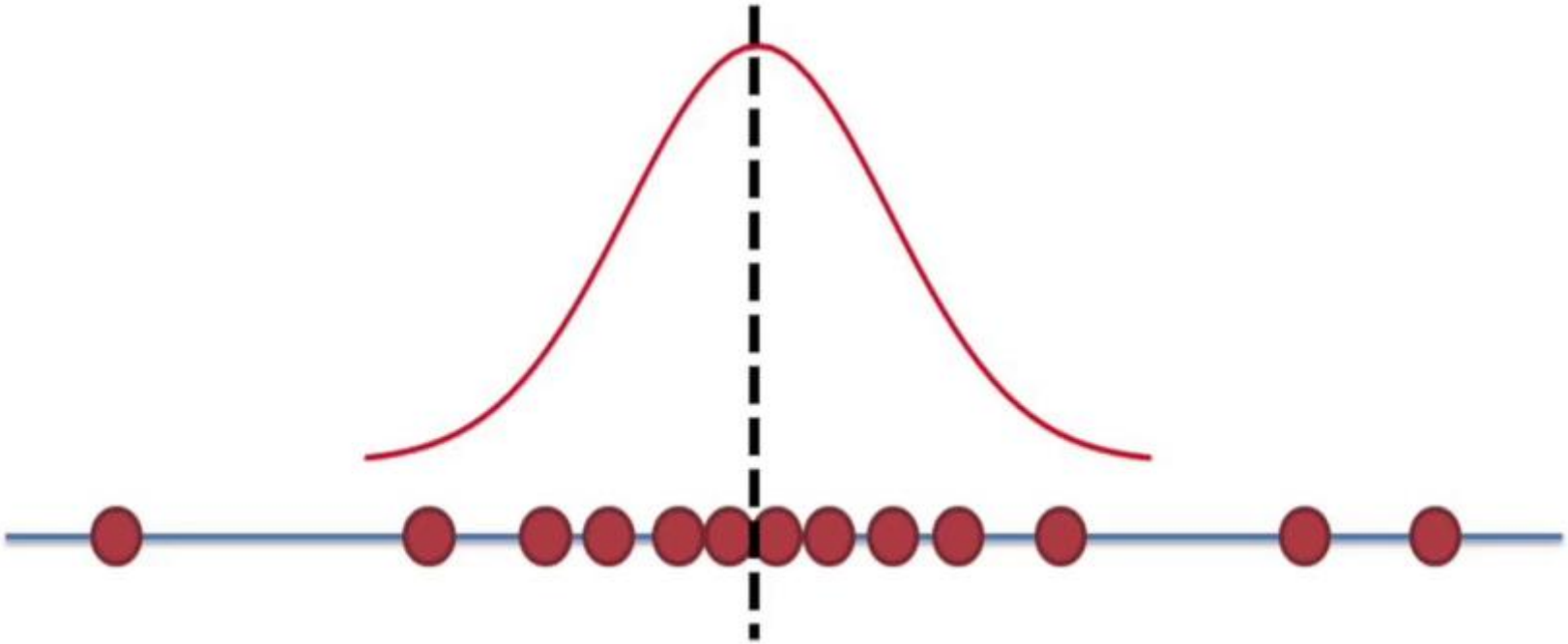
Is one location “better” than another?

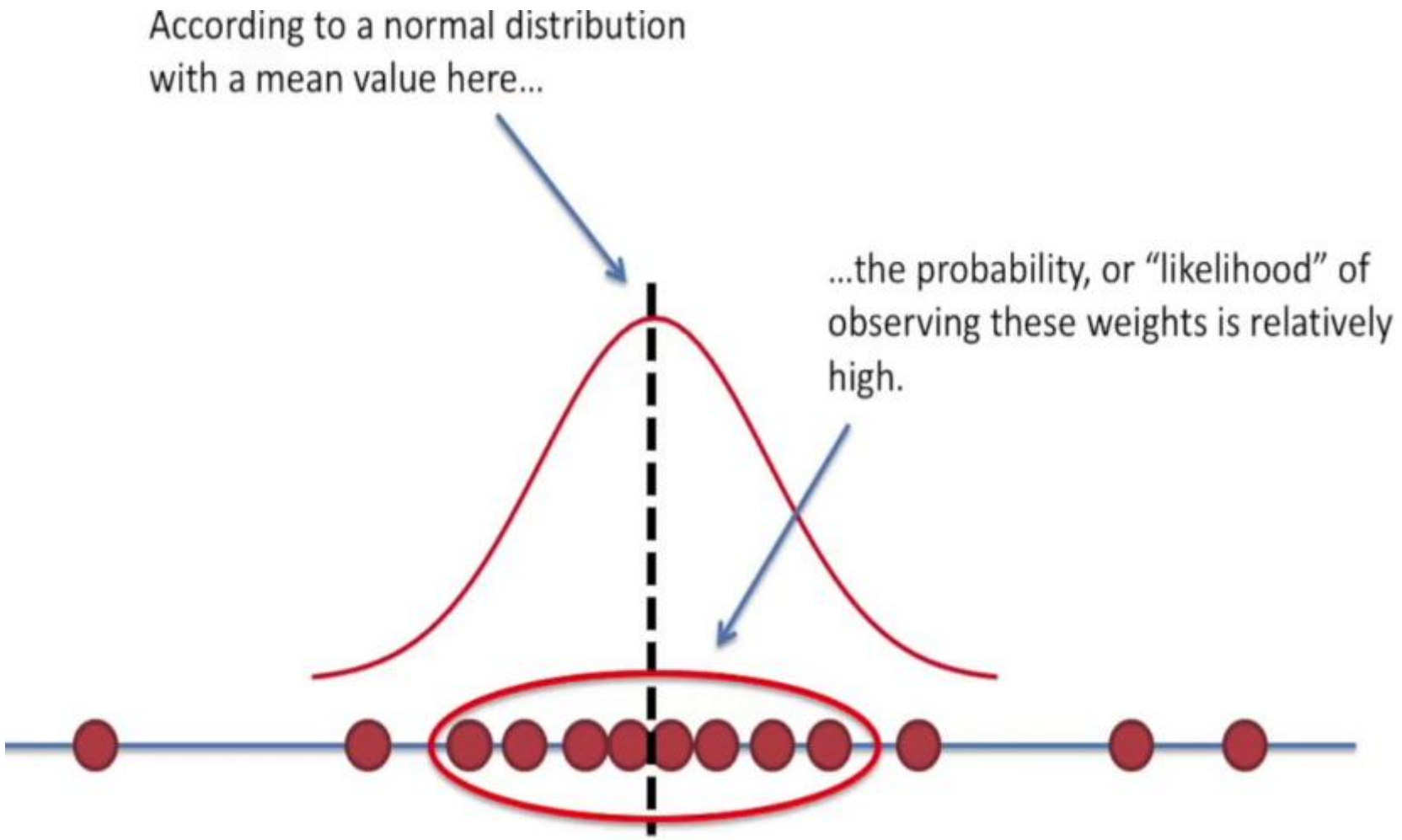




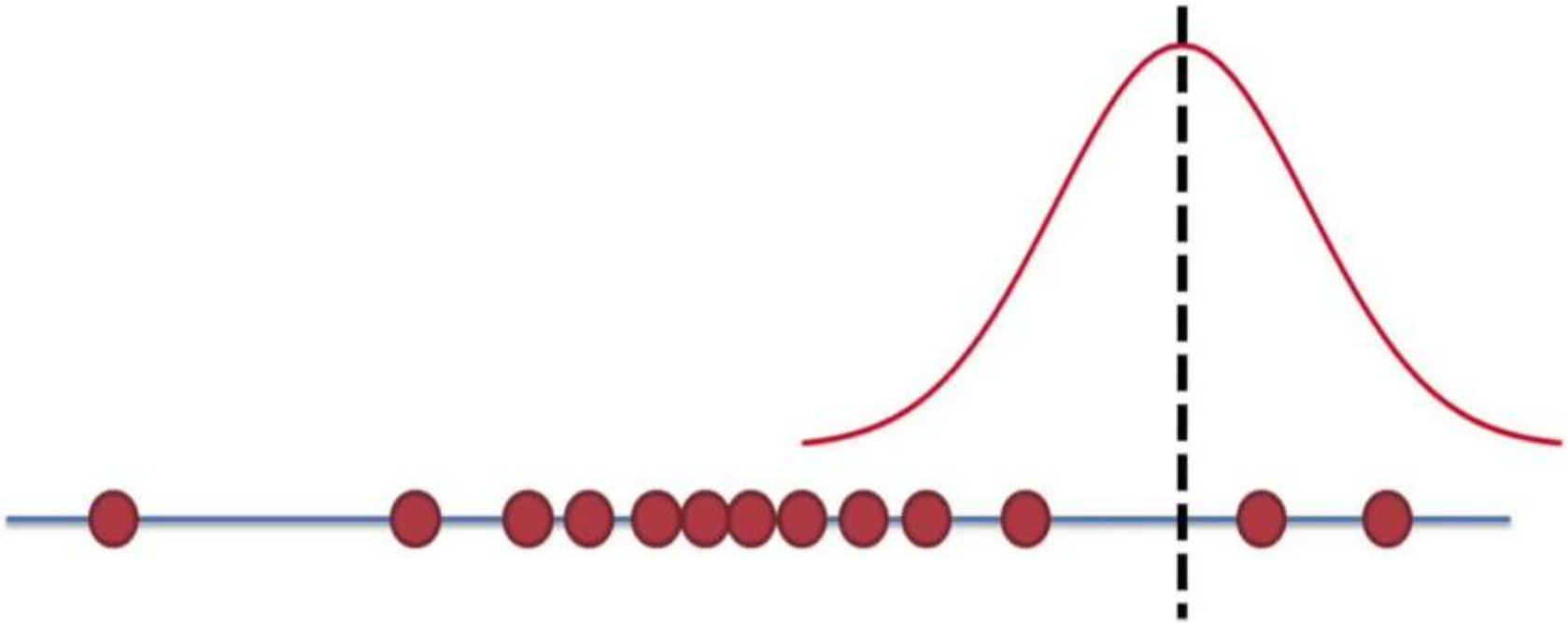

According to a normal distribution with a mean value over here...
...the probability, or "likelihood" of observing all these weights is low.

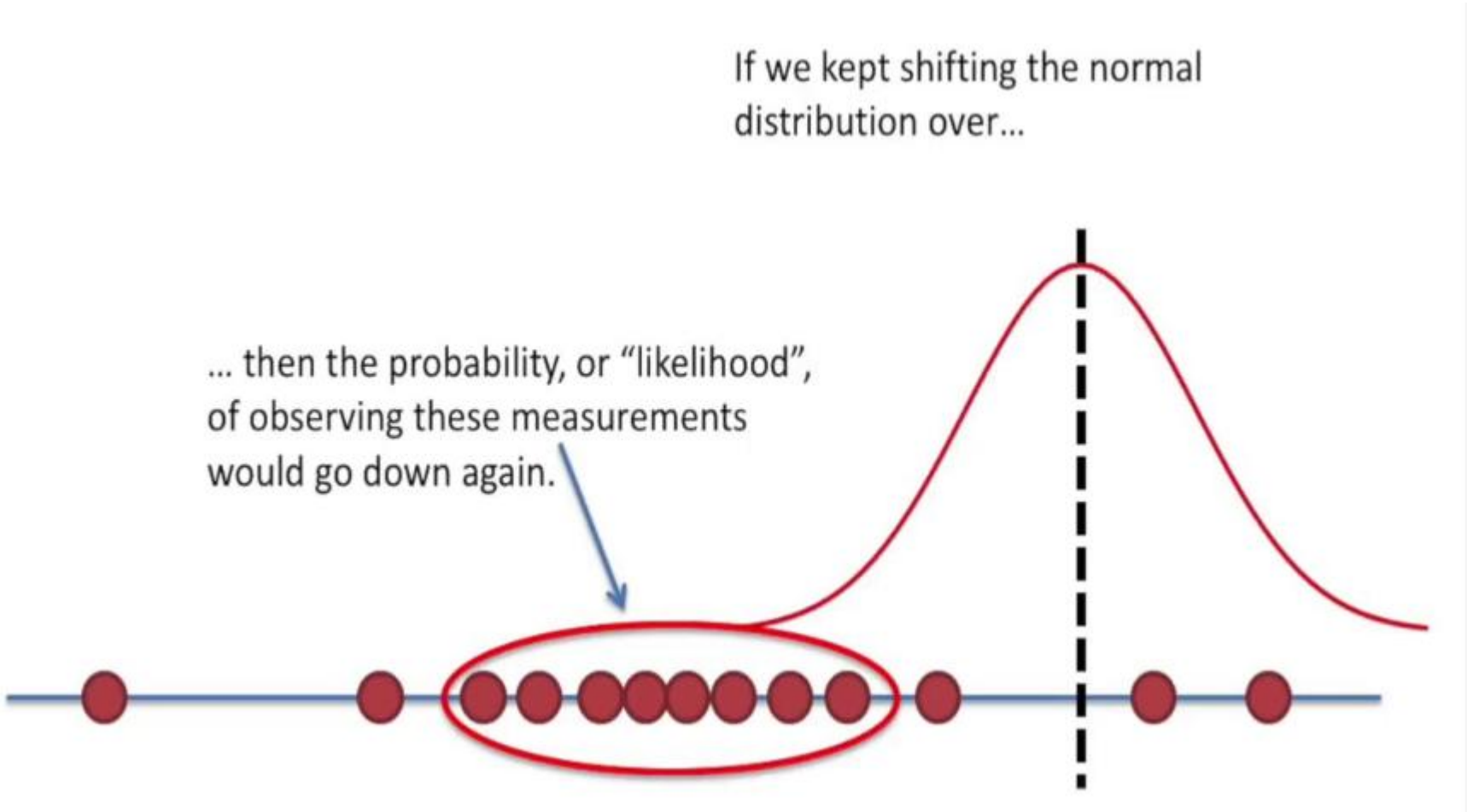
What if we shifted the normal distribution over, so that its mean was the same as the average weight?




According to a normal distribution
with a mean value here...
...the probability, or "likelihood" of
observing these weights is relatively
high.

If we kept shifting the normal distribution over…




If we kept shifting the normal distribution over…
… then the probability, or "likelihood", of observing these measurements would go down again.

می خواهیم تابع چگالی P(x) را به دست آوریم.

یک تابع چگالی احتمالی پارامتری برای P(x) تعریف میکنیم.

با فرض وجود مشاهده های $X = (x^1 , \dots , x^n)$ سعی می کنیم پارامترهای مدل را بهینه کنیم تا احتمال P(x) بیشینه گردد.

$$X = (x^1 , \dots , x^n)$$

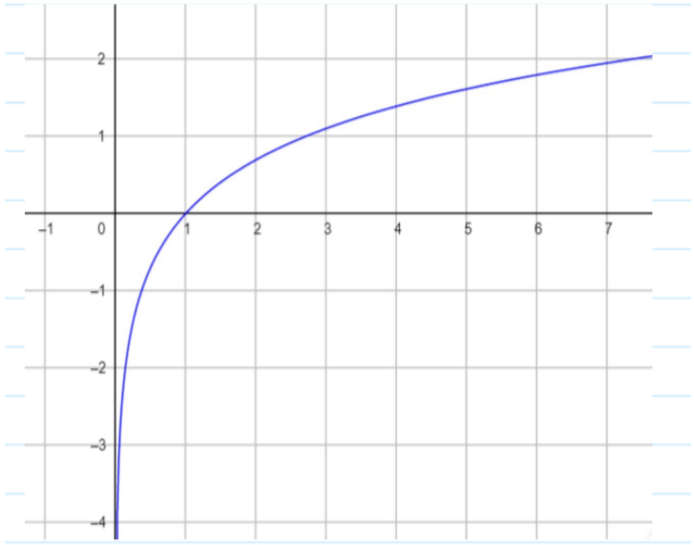
فرض می کنیم هر داده از توزیع $P(x \mid \theta)$ به صورت مستقل به دست آمده اند.

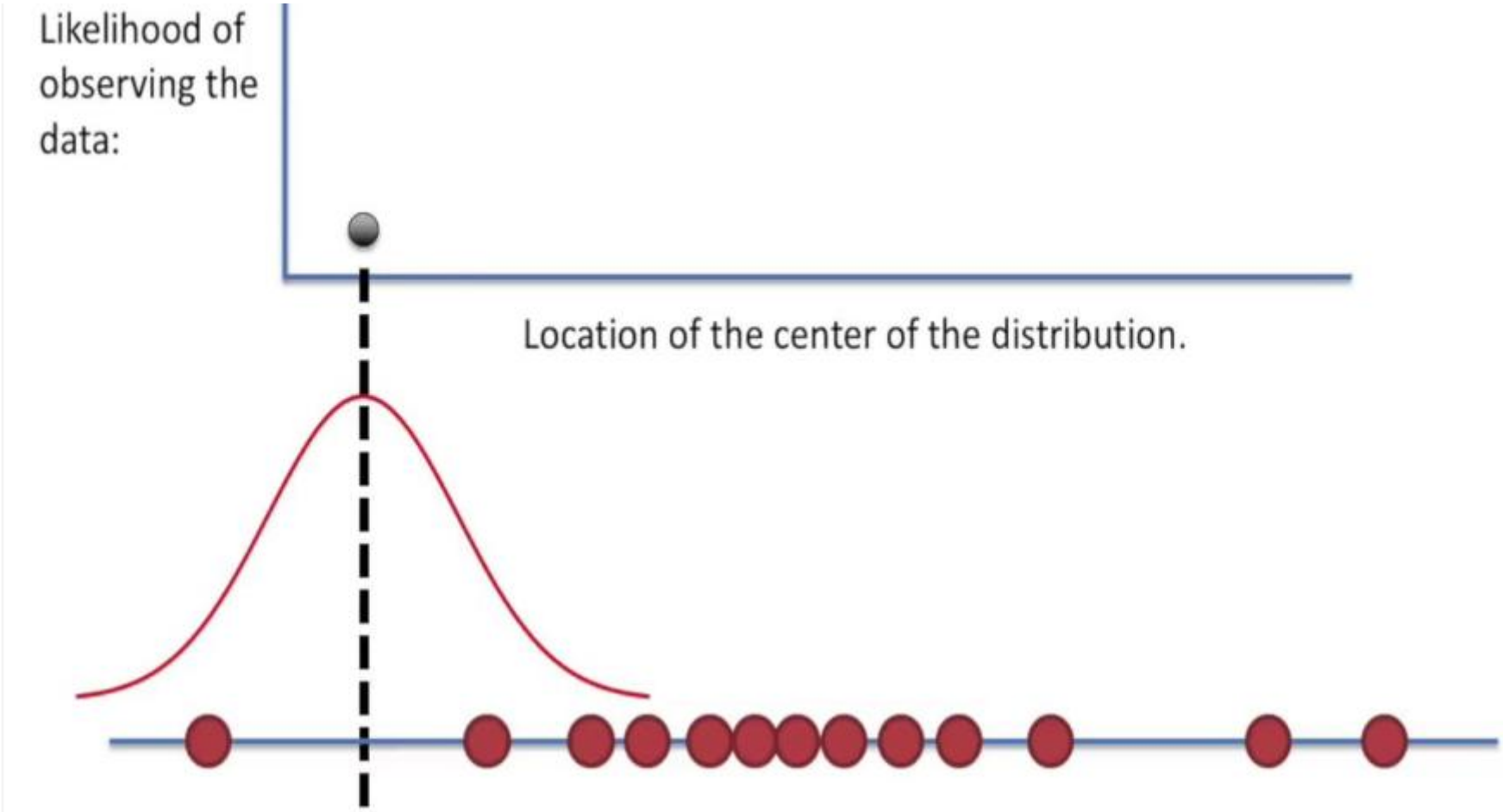
$$P(X \mid \theta) = \prod_{n=1}^{N} P(x^n|\theta) = L(\theta)$$

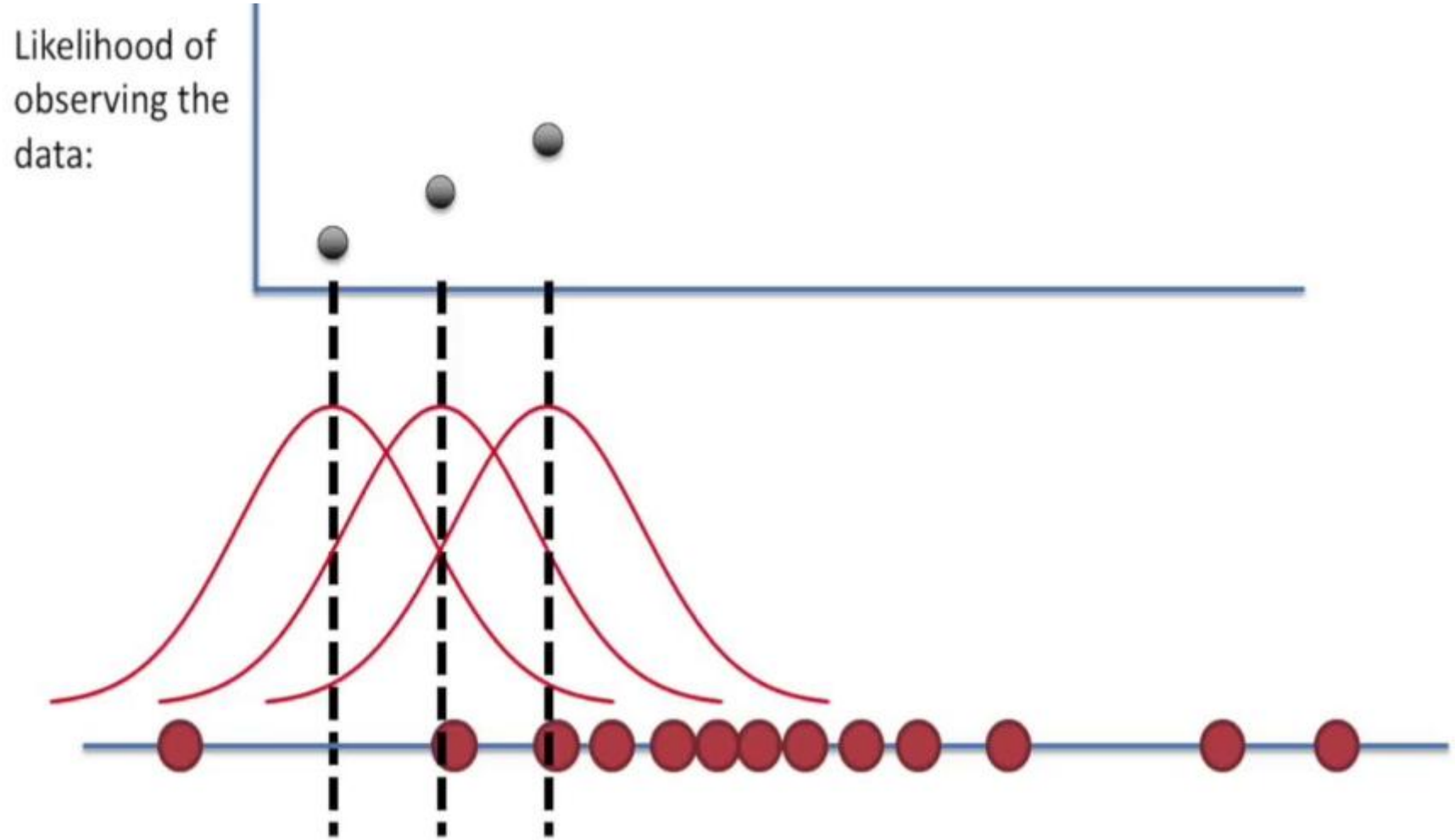
$$\hat{\theta} = \underset{\theta}{\operatorname{argmax}}\, L(\theta)$$

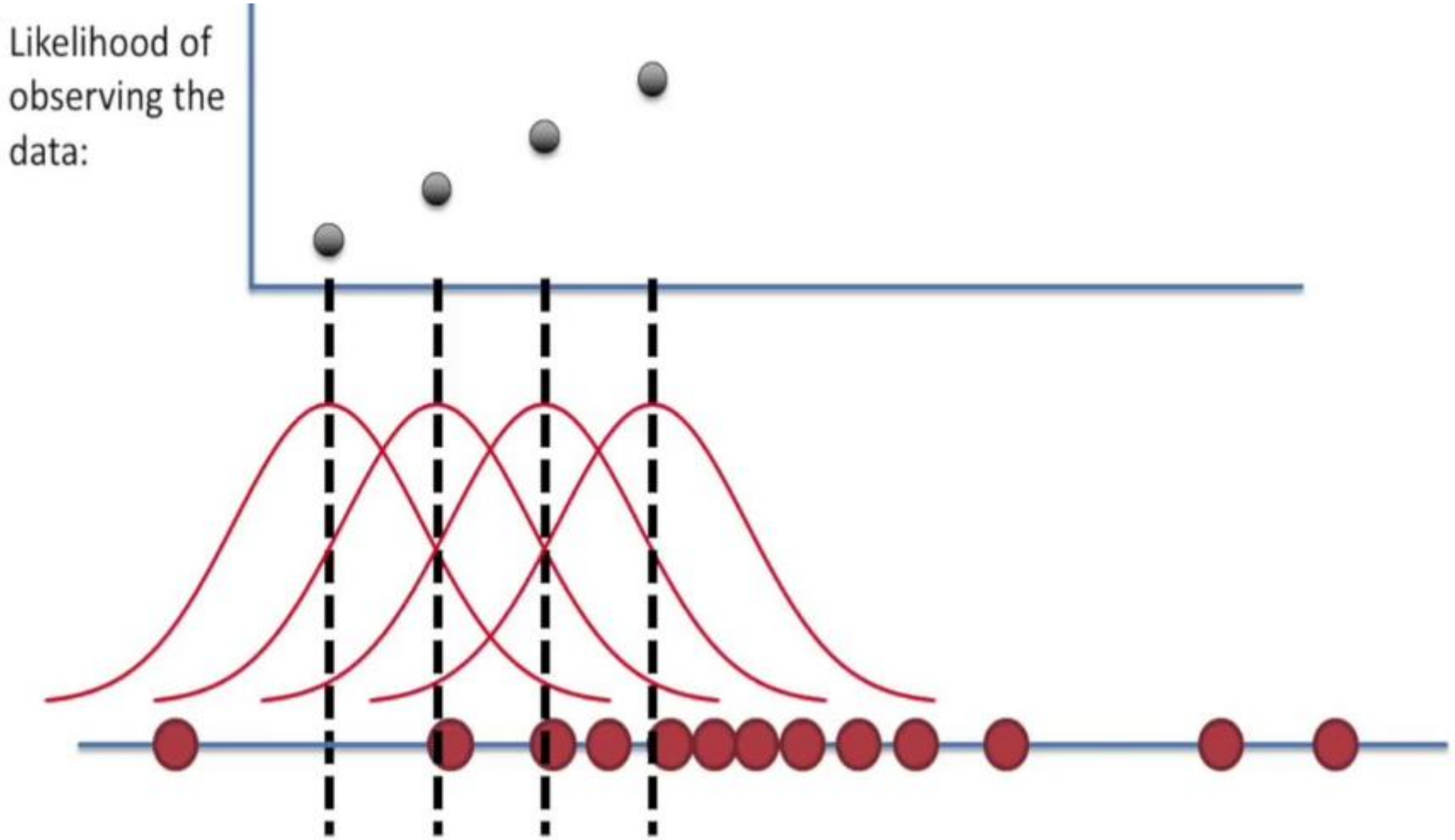
$$\text{Log } P(X \mid \theta) = \sum_{n=1}^{N} \log P(x^n \mid \theta) = \log L(\theta)$$

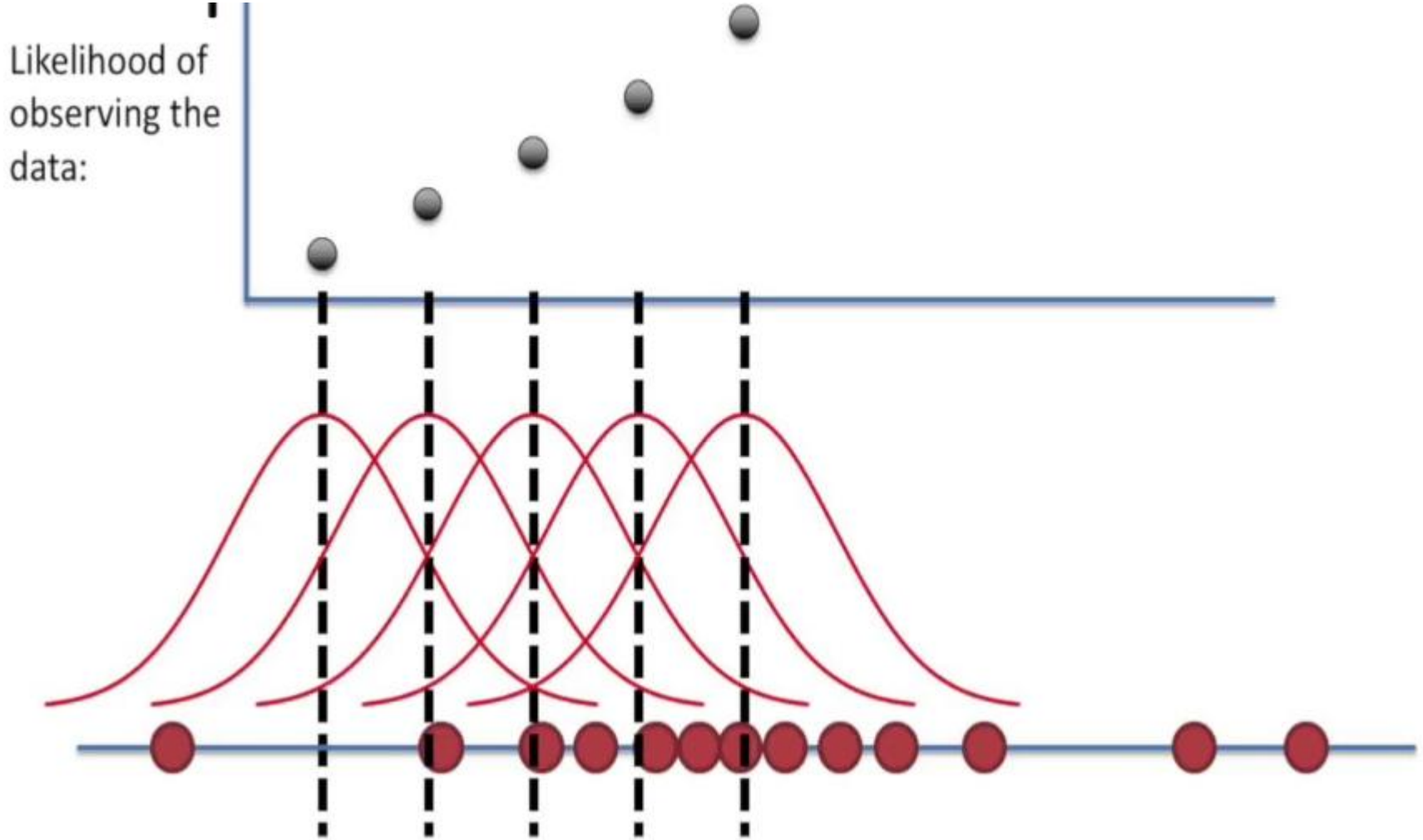
$$\hat{\theta} = \underset{\theta}{\operatorname{argmax}}\, L(\theta) = \underset{\theta}{\operatorname{argmax}}\, \log L(\theta)$$

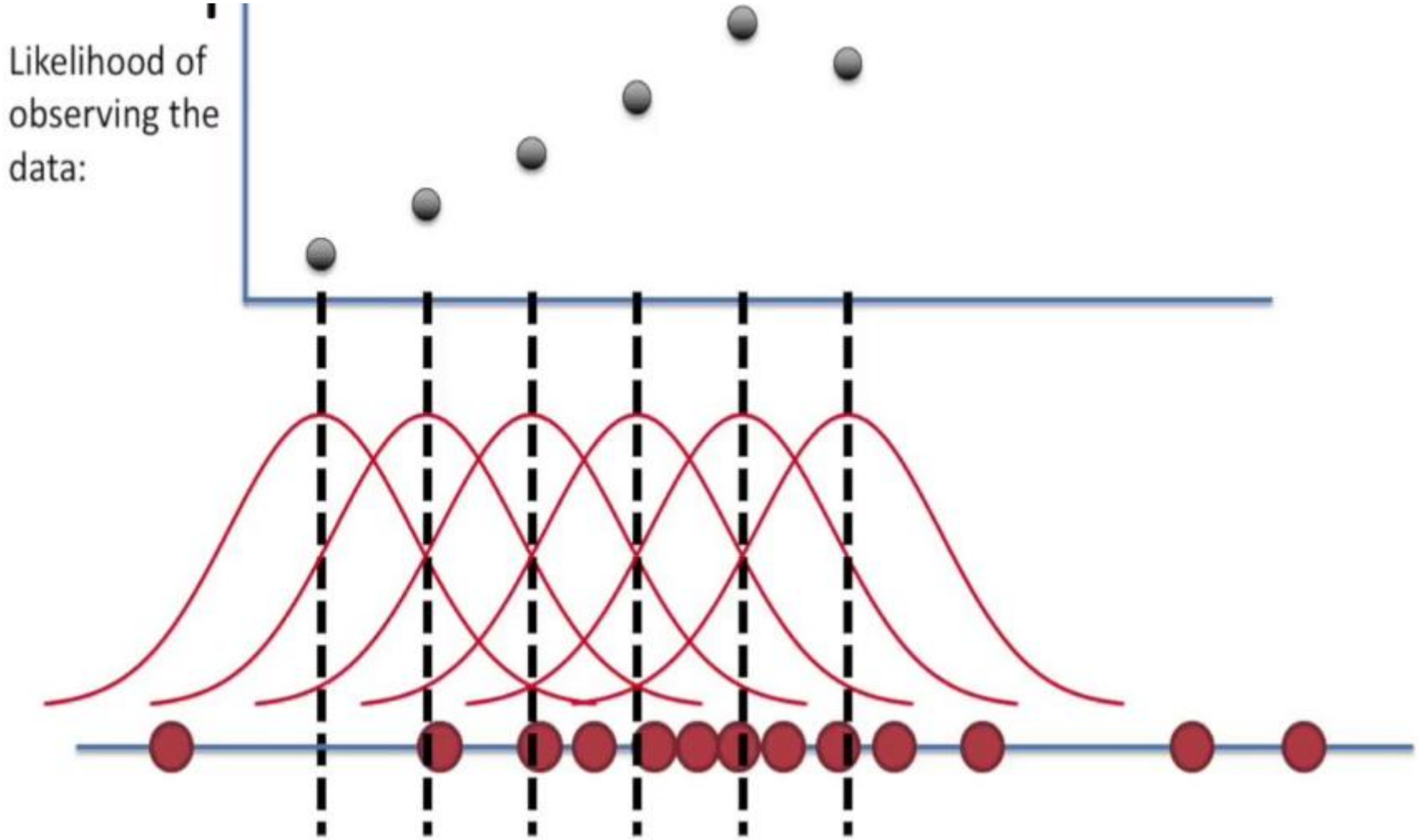


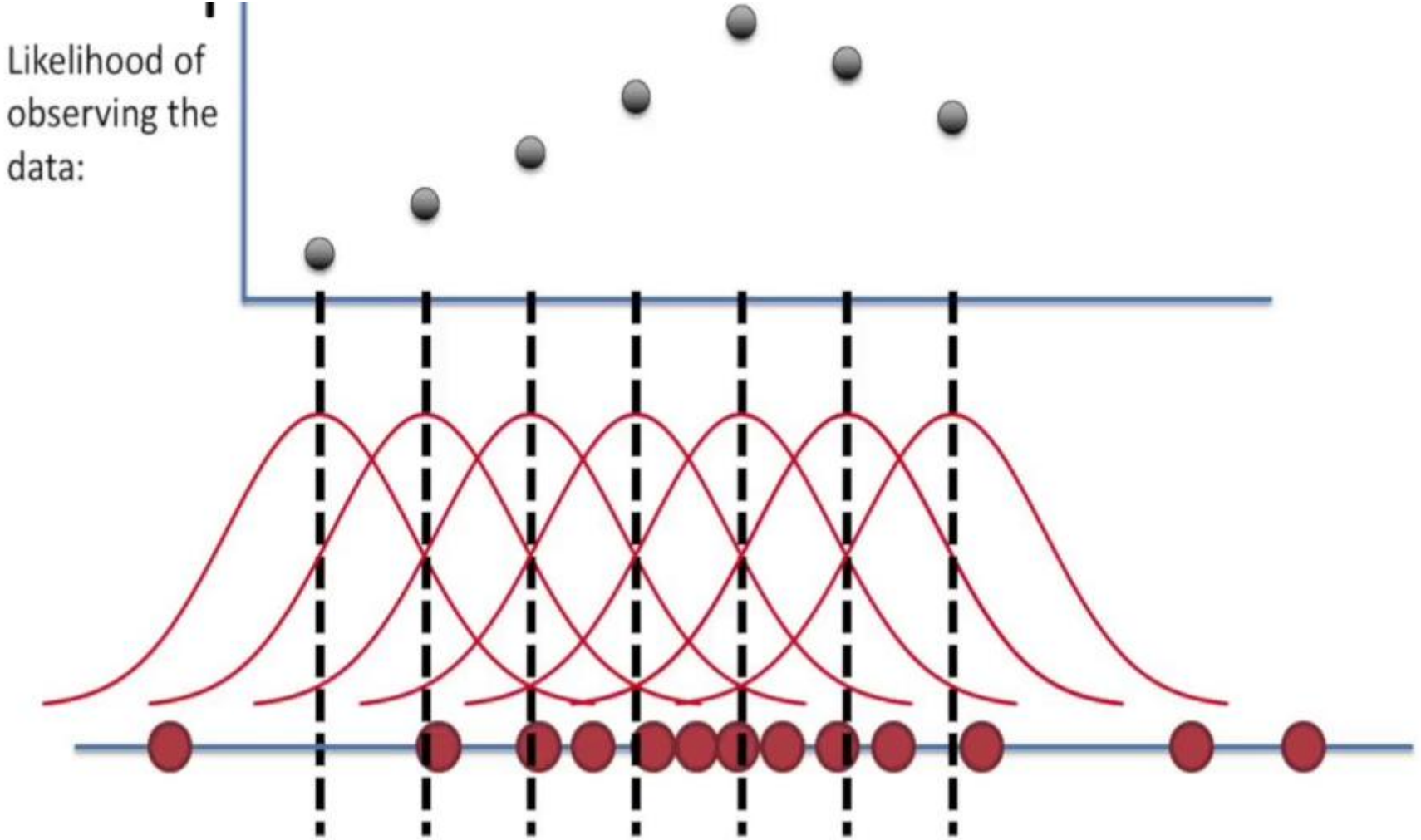

Likelihood of observing the data:
Location of the center of the distribution.

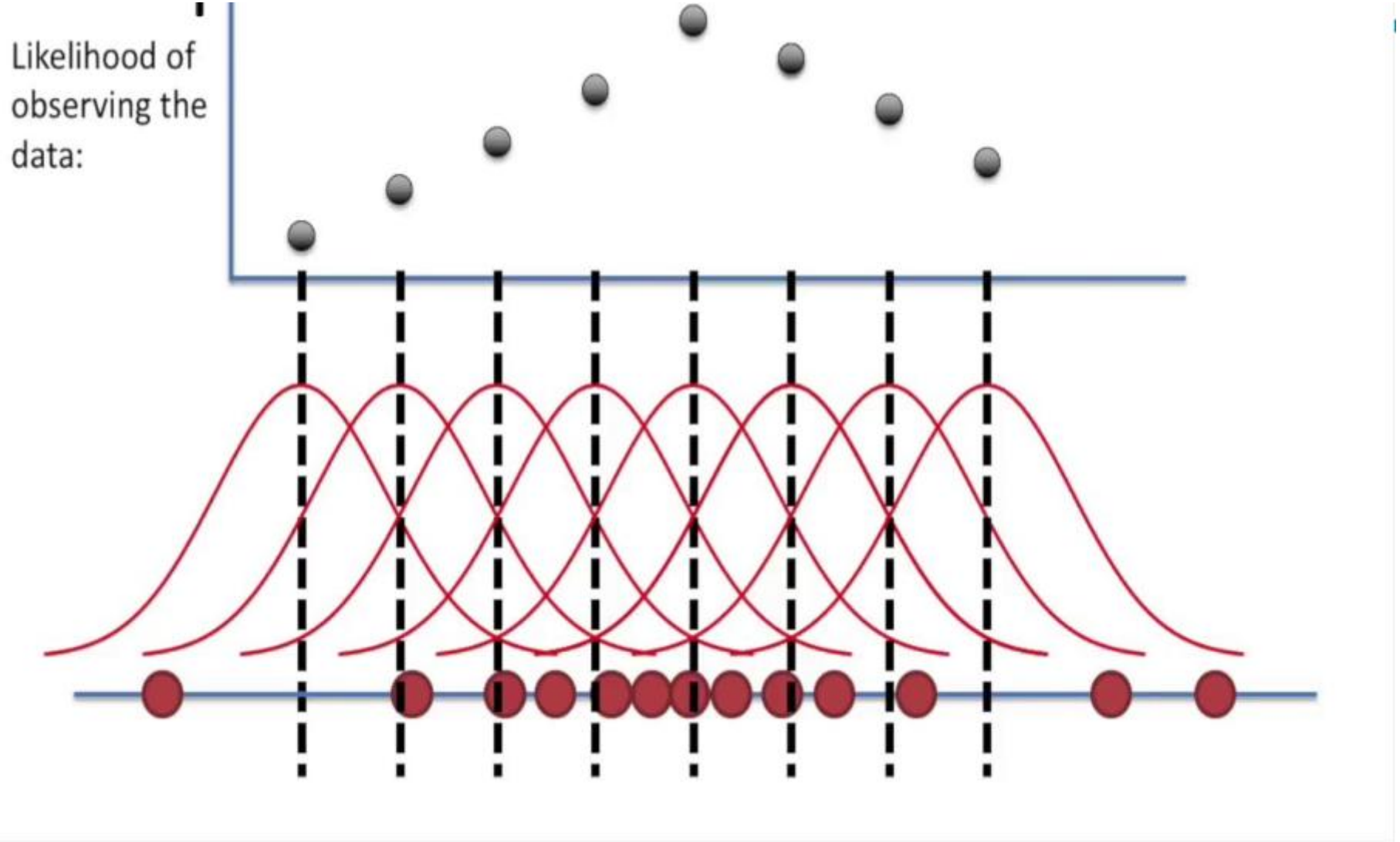

Likelihood of
observing the
data:


Likelihood of
observing the
data:


Likelihood of
observing the
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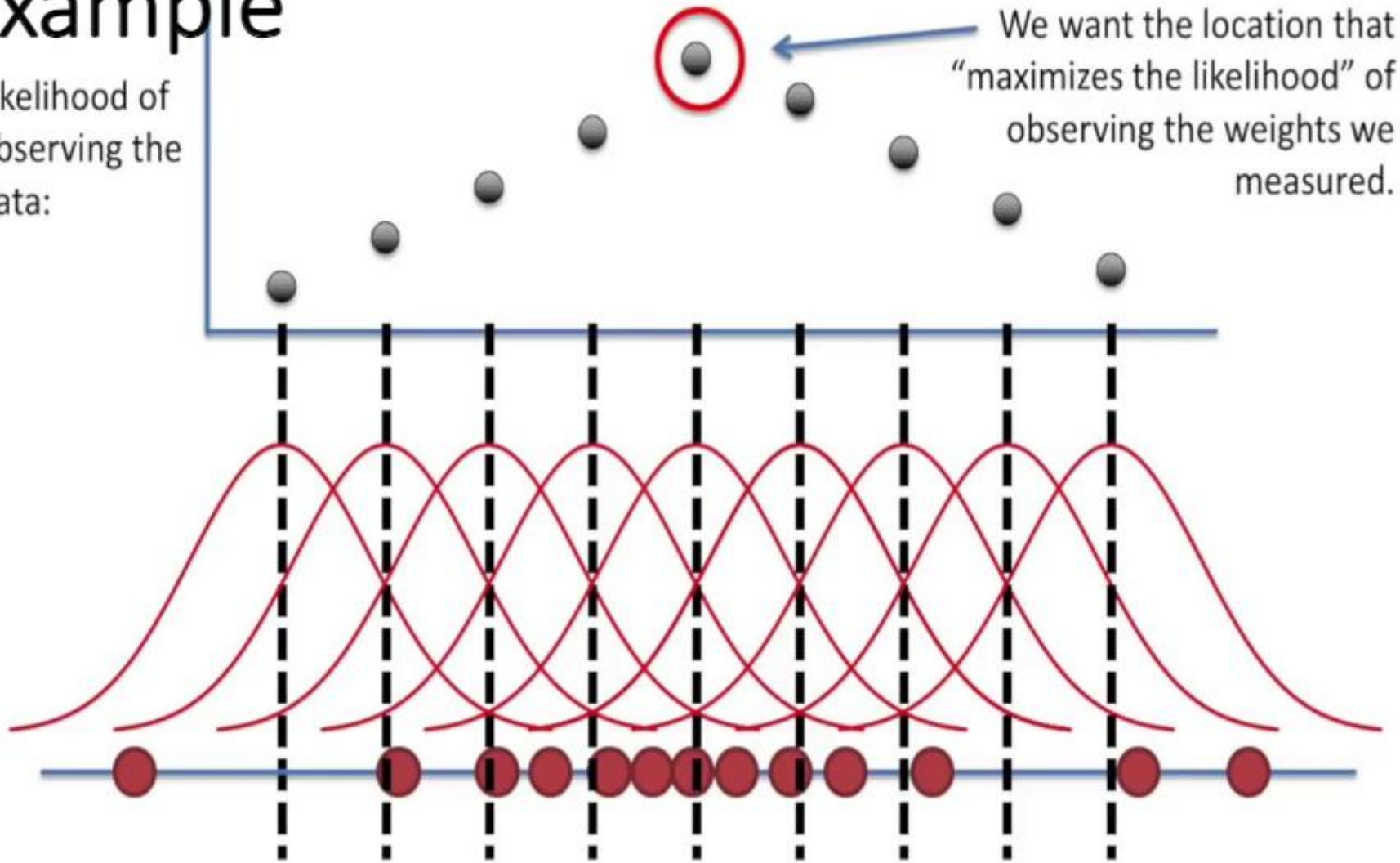

Likelihood of
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data:


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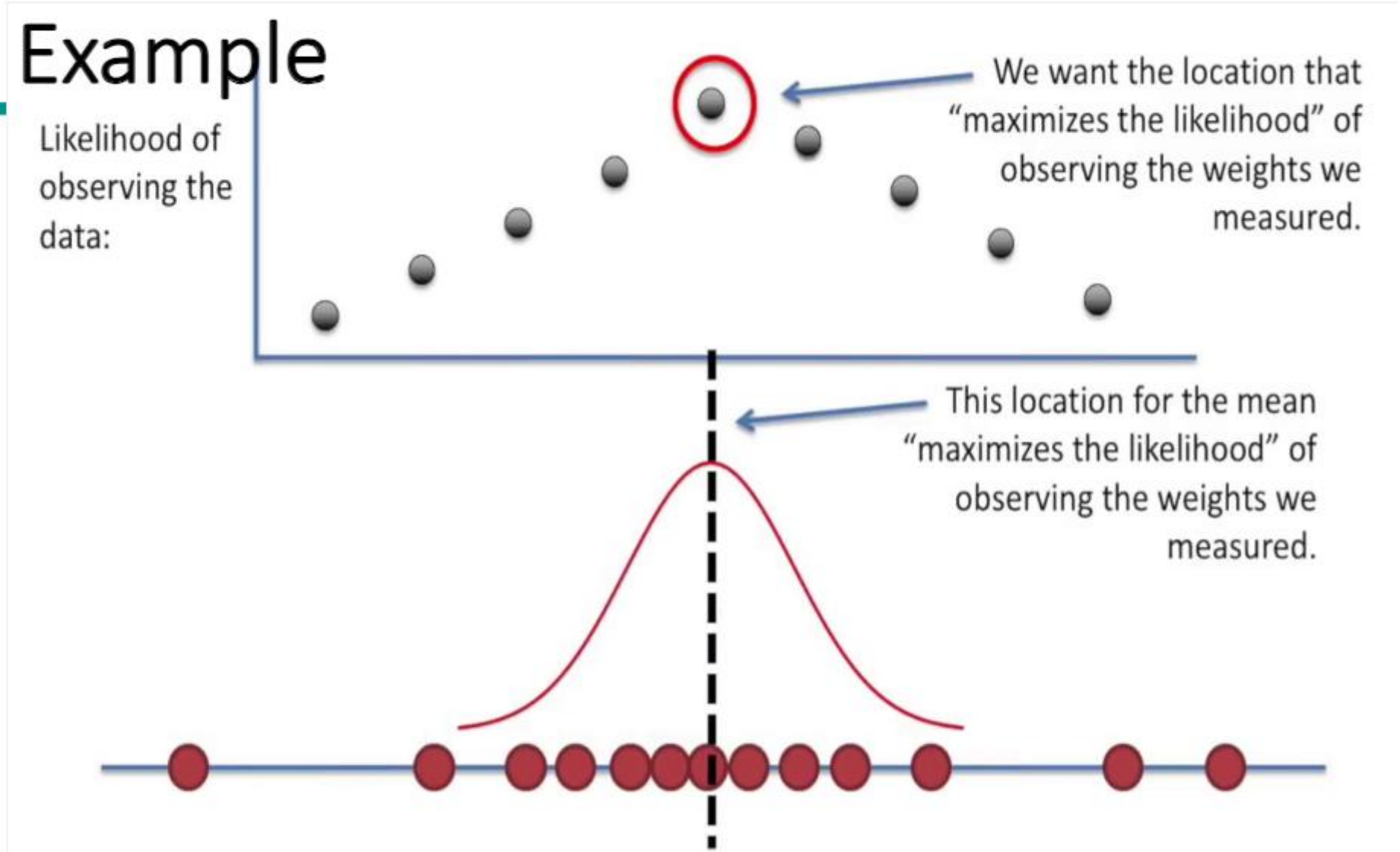

Likelihood of
observing the
data:

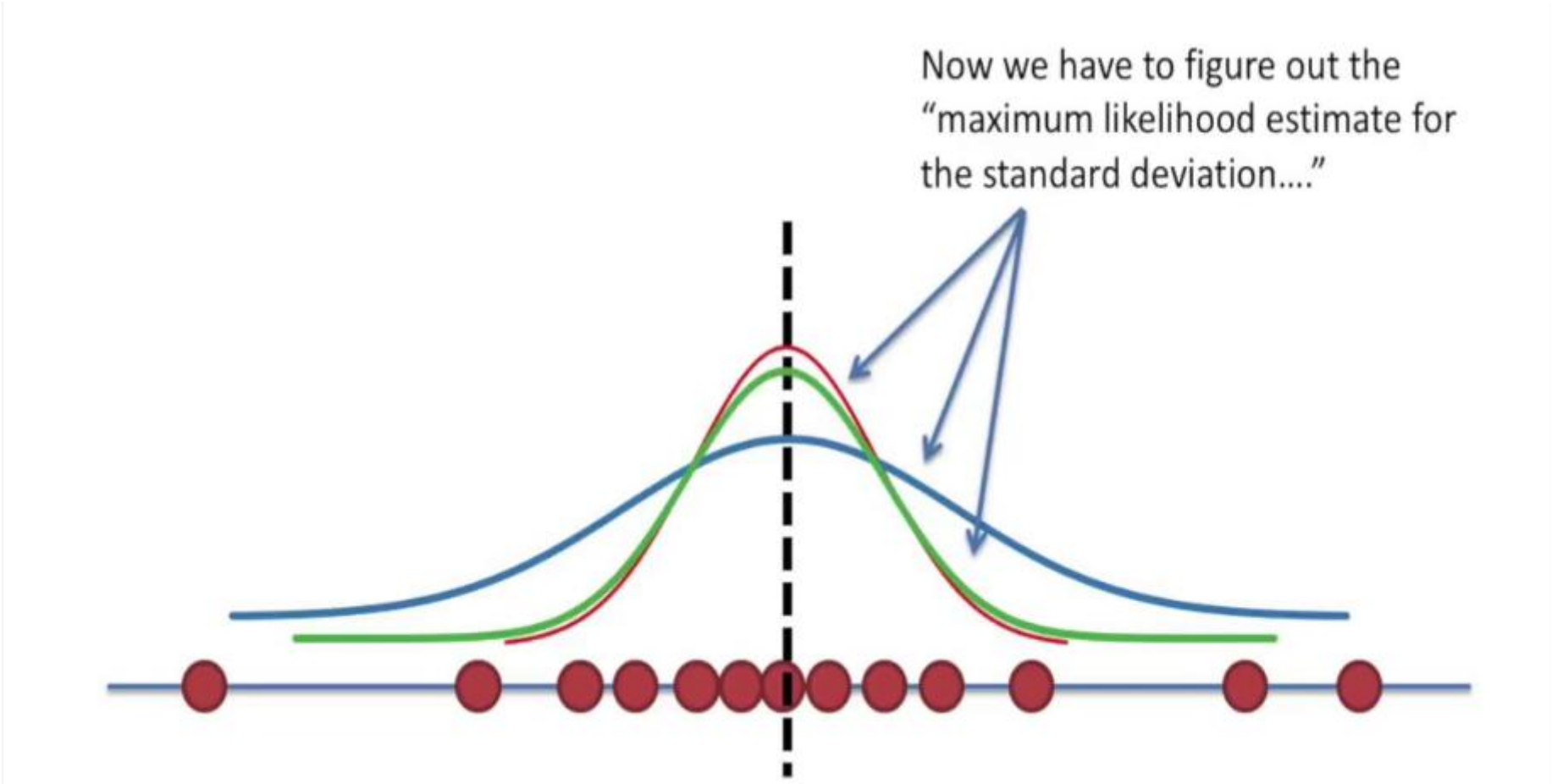
MLE Example

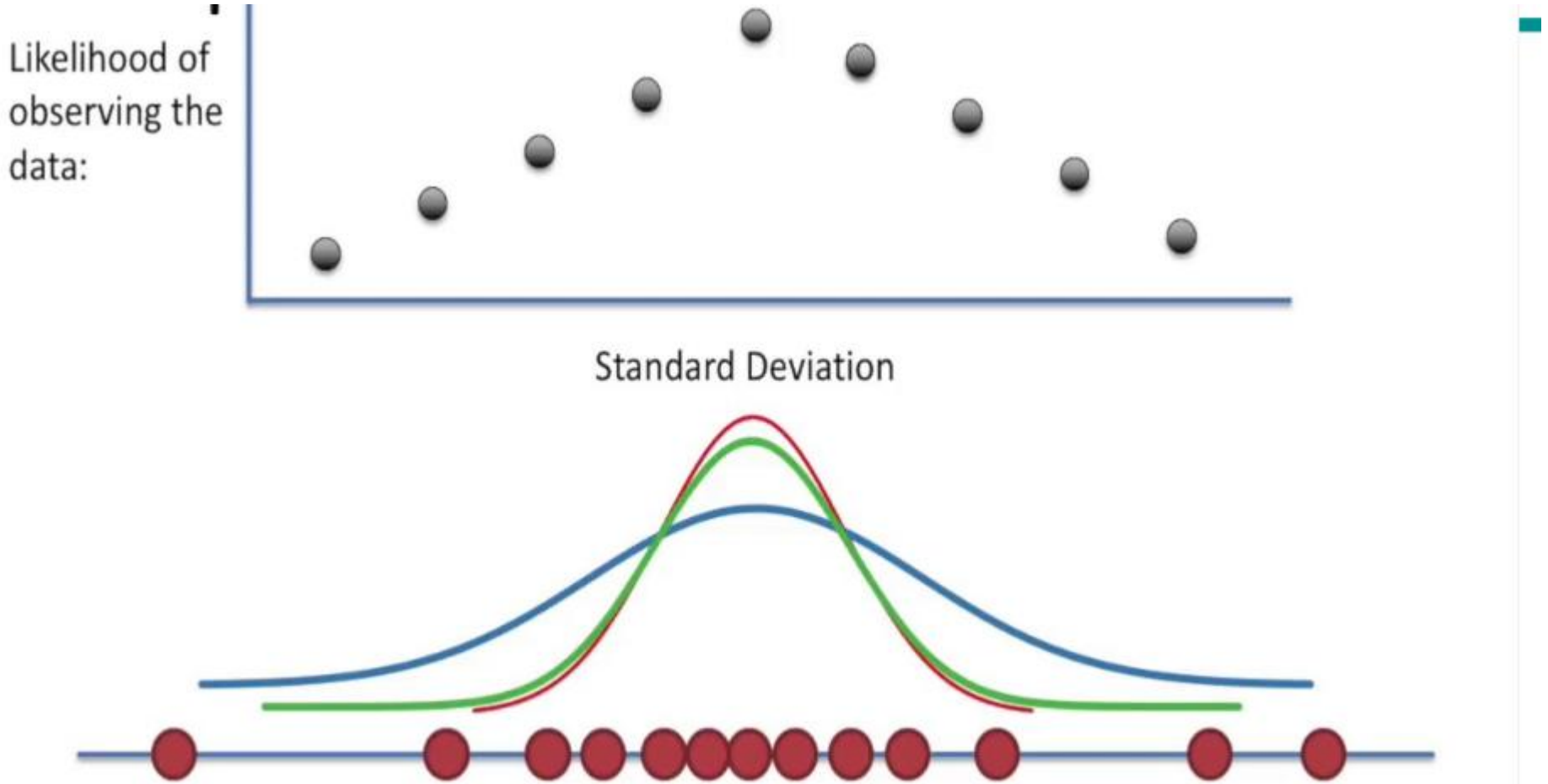
Likelihood of observing the data:



MLE Example

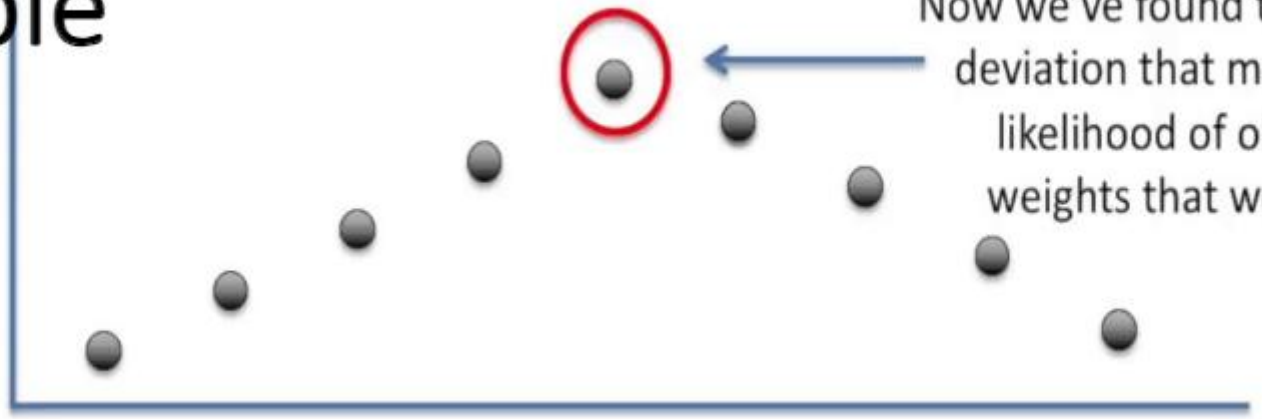

Likelihood of observing the data:
We want the location that "maximizes the likelihood" of observing the weights we measured.
This location for the mean "maximizes the likelihood" of observing the weights we measured.


Now we have to figure out the
“maximum likelihood estimate for
the standard deviation….”


Likelihood of observing the data:
Standard Deviation

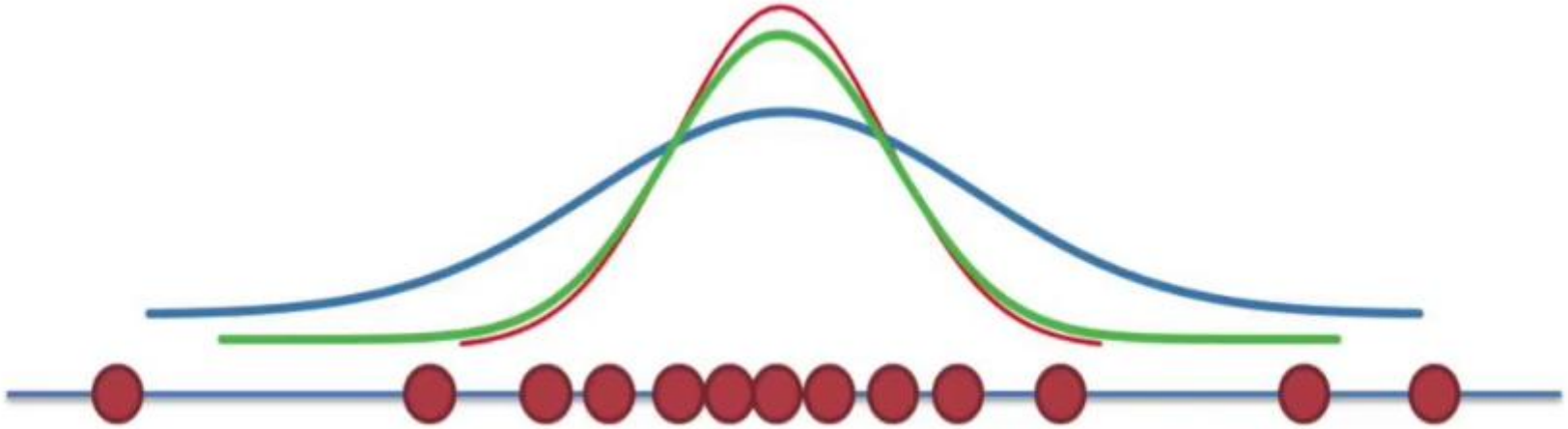
MLE Example

Likelihood of observing the data:

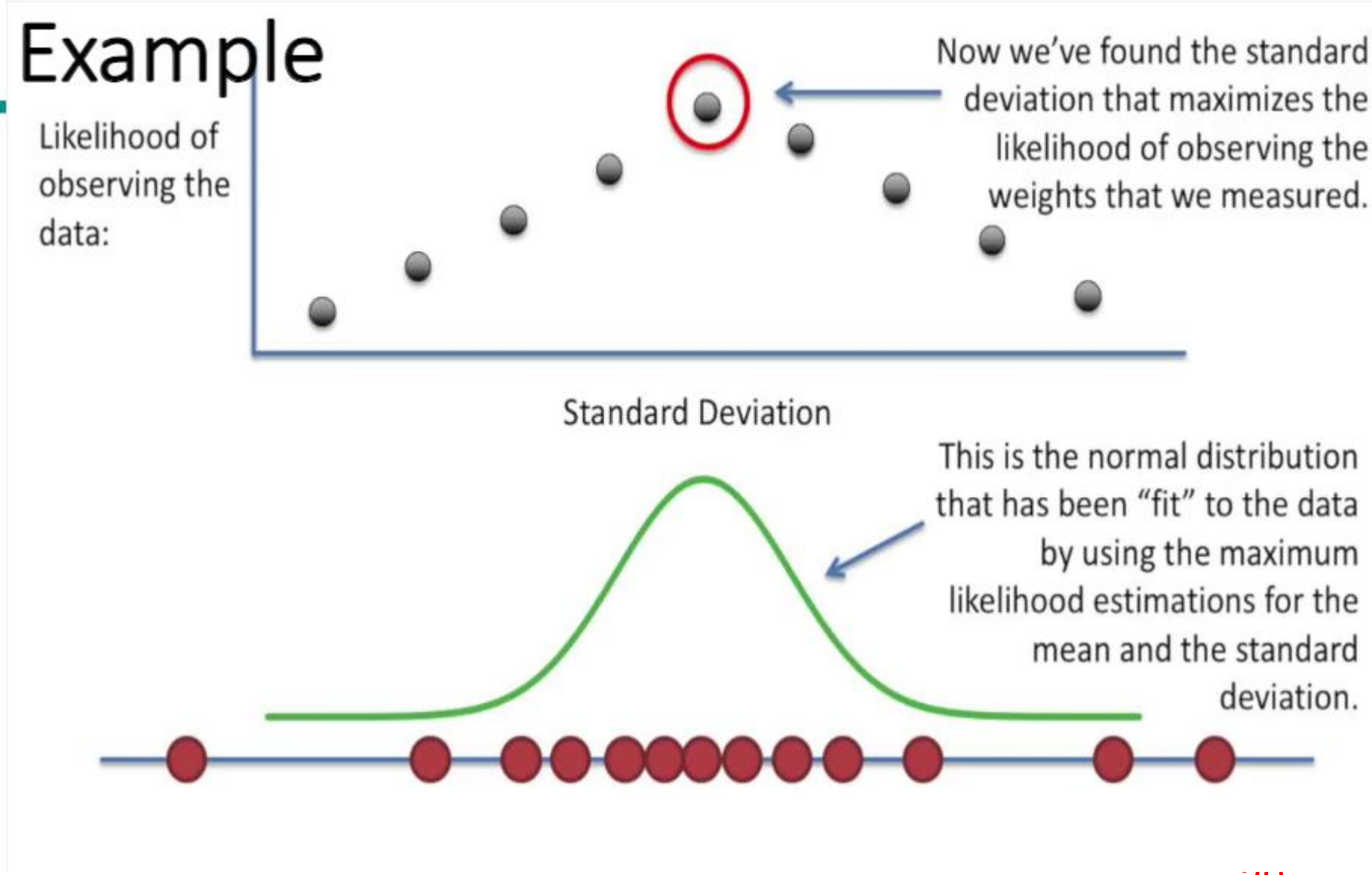


Now we've found the standard deviation that maximizes the likelihood of observing the weights that we measured.

Standard Deviation



MLE Example


Likelihood of observing the data:
Now we've found the standard deviation that maximizes the likelihood of observing the weights that we measured.
Standard Deviation
This is the normal distribution that has been "fit" to the data by using the maximum likelihood estimations for the mean and the standard deviation.

Calculating the MLE

- Probability of observing a single data point *x*

$$P(x;\mu,\sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Parameters

- Example: $P(9, 9.5, 11;\mu,\sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(9-\mu)^2}{2\sigma^2}\right) \times \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(9.5-\mu)^2}{2\sigma^2}\right) \times \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(11-\mu)^2}{2\sigma^2}\right)$





The Log likelihood

- Maximum is found by differentiation, i.e., find the derivative of the function w.r.t. a variable, set it to zero and find the required value.
- Since the previous expression is not easy to differentiate, we simplify the calculus considering the natural logarithm of the expression.

$$\ln(P(x;\mu,\sigma)) = \ln\left(\frac{1}{\sigma\sqrt{2\pi}}\right) - \frac{(9-\mu)^2}{2\sigma^2} + \ln\left(\frac{1}{\sigma\sqrt{2\pi}}\right) - \frac{(9.5-\mu)^2}{2\sigma^2} + \ln\left(\frac{1}{\sigma\sqrt{2\pi}}\right) - \frac{(11-\mu)^2}{2\sigma^2}$$

$$\ln(P(x;\mu,\sigma)) = -3\ln(\sigma) - \frac{3}{2}\ln(2\pi) - \frac{1}{2\sigma^2}\left[(9-\mu)^2 + (9.5-\mu)^2 + (11-\mu)^2\right]$$

Derivation with respect to mu

- This expression can be easily differentiated to find the maximum.

$$\frac{\partial \ln(P(x;\mu,\sigma))}{\partial \mu} = \frac{1}{\sigma^2}\left[9 + 9.5 + 11 - 3\mu\right].$$

$$\mu = \frac{9 + 9.5 + 11}{3} = 9.833$$

- The same can be done for the standard deviation.





Let $x_1, x_2, \dots, x_n$ be a random sample from a normal distribution with unknown mean μ and variance σ^2 .

Find Maximum Likelihood estimators of mean μ and variance σ^2 .

Answer

In finding the estimators , the first thing we will do is write the probability density function as a function of $\theta_1 = \mu$ and $\theta_2 = \sigma^2$

$$f(x_i ; \theta_1 , \theta_2) = \frac{1}{\sqrt{\theta_2}\ \sqrt{2\pi}} \exp\left[\frac{-(x_i-\theta_1)^2}{2\theta_2}\right]$$

For $-\infty < \theta_1 < \infty$ and $0 < \theta_2 < \infty$. We do this so as not to cause confusion when taking the derivative of the likelihood with respect to σ^2. Now , that makes the likelihood function:

$$L(\theta_1 , \theta_2) = \prod_{i=1}^{n} f(x_i ; \theta_1 , \theta_2) = \theta_2^{-n/2}\ (2\pi)^{-n/2} \exp\left[\frac{-1}{2\theta_2} \sum_{i=1}^{n}(x_i - \theta_1)^2\right]$$





And therefore the log of the likelihood function:

$$\text{Log L}(\theta_1 , \theta_2) = \frac{-n}{2} \log \theta_2 - \frac{n}{2} \log (2\pi) - \frac{\sum(x_i - \theta_1)^2}{2\theta_2}$$

Now, upon taking the partial derivative of the log likelihood with respect to θ_1 , and setting to 0 , we see that a few things cancel each other out , leaving us with:

$$\frac{\partial \text{ Log L}(\theta_1 , \theta_2)}{\partial \theta_1} = \frac{-2 \sum(x_i - \theta_1)(-1)}{2\theta_2} \equiv 0$$

Now, multiplying through by θ_2 and distributing the summation , we get:

$$\sum(x_i) - n\theta_1 = 0$$

Now , solving for θ_1 and putting on its hat we have shown that the maximum likelihood estimate of θ_1 is :

$$\hat{\theta}_1 = \hat{\mu} = \frac{\sum x_i}{n} = \bar{x}$$

Now for θ_2 taking the partial derivative of the log likelihood with respect to θ_2 , and setting to 0 , we get:

$$\frac{\partial \text{ Log } L(\theta_1 , \theta_2)}{\partial \theta_2} = \frac{-n}{2\theta_2} + \frac{\sum (x_i - \theta_1)^2}{2\theta_2^2} = 0$$

Multiplying through by $2\theta_2^2$:

$$\frac{\partial \text{ Log } L(\theta_1 , \theta_2)}{\partial \theta_2} = [\frac{-n}{2\theta_2} + \frac{\sum (x_i - \theta_1)^2}{2\theta_2^2} = 0] * 2\theta_2^2$$





We get:

$$-n\theta_2 + \sum(x_i - \theta_1)^2 = 0$$

And , solving for θ_2 , and putting on its hat , we have shown that the maximum likelihood estimate of θ_2 is:

$$\hat{\theta}_2 = \hat{\sigma}^2 = \frac{\sum(x_i - \bar{x})^2}{n}$$





ارتباط روش LS , MLE

Least square (LS)

یک تابع هزینه تعریف کردیم و با توجه به داده ها مدلی را پیدا کردیم که تابع هزینه را کمینه می کرد.

در این قسمت یک نگاه جدید داریم و می خواهیم از منظر مدل های احتمالاتی به این مسئله نگاه کنیم و به عبارتی یک تعبیر احتمالاتی از مسئله LS داشته باشیم.





یک مدل احتمالاتی برای LS

فرض کنید داده های ما توسط مدل زیر تولید می شوند:

$$y_n = x_n^T \theta + \varepsilon_n$$

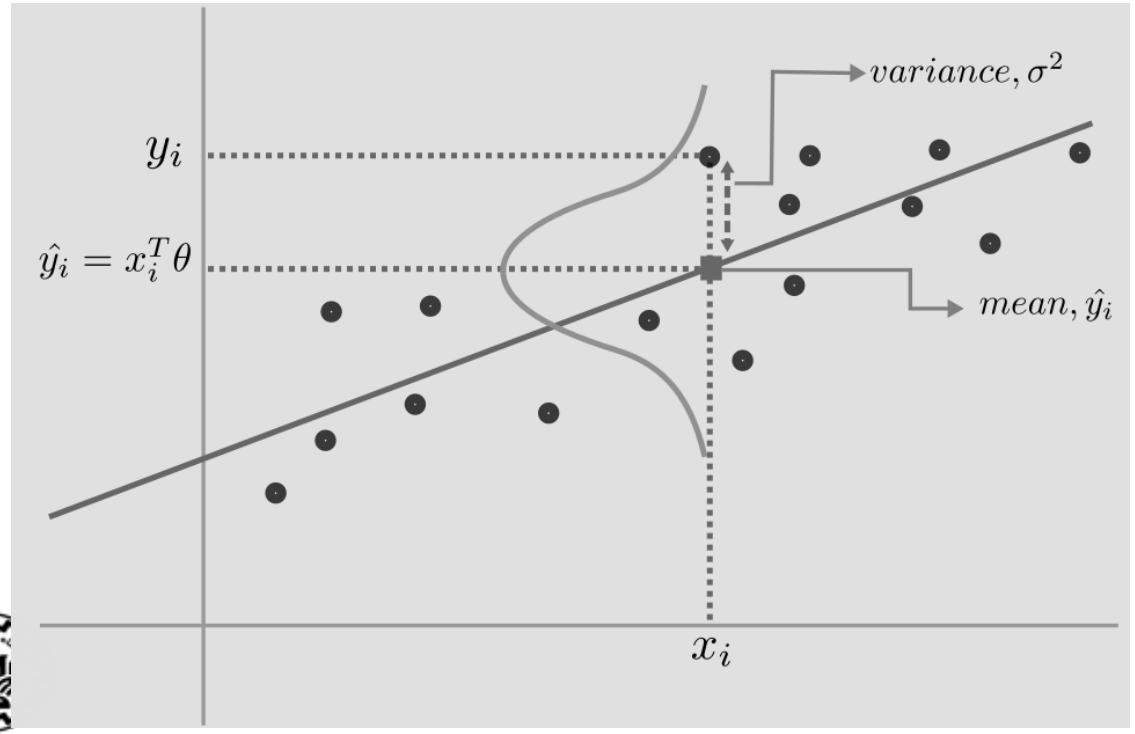
$\varepsilon_n \sim \mathrm{N}(\mu\ , \sigma^2)$:

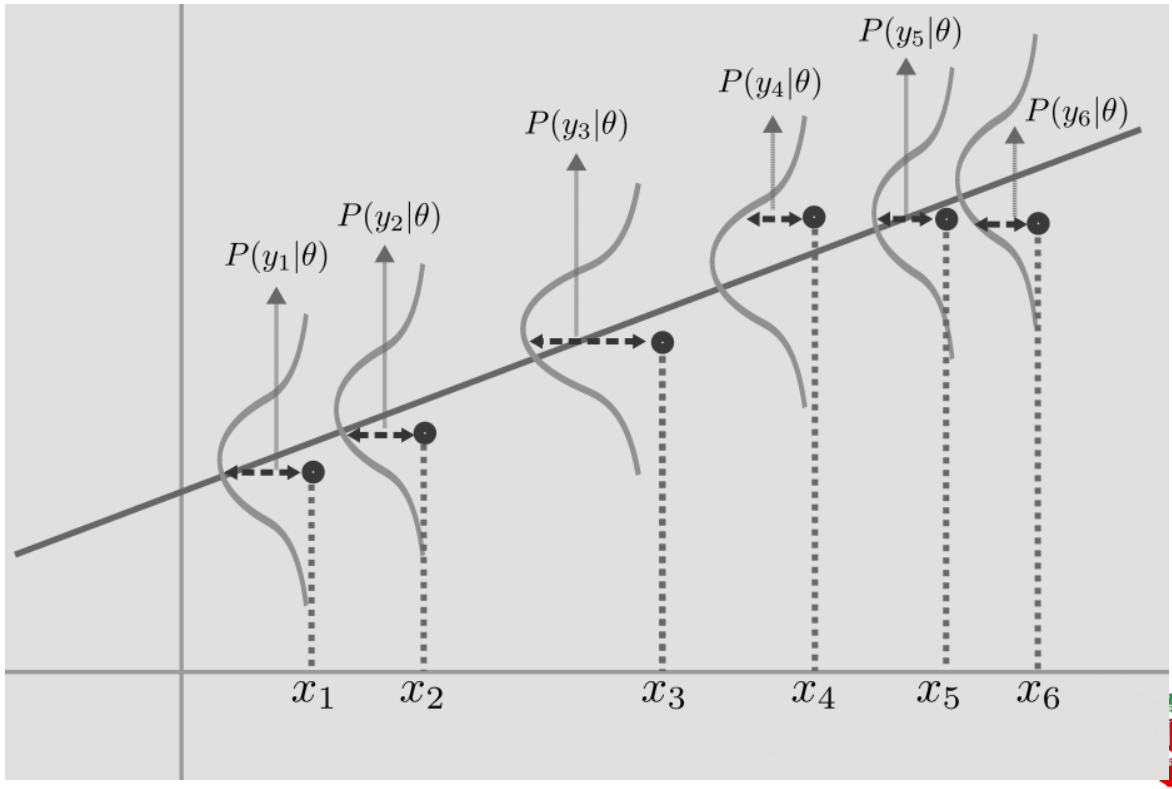
ε_n یک نویز گاوسی با میانگین صفر و واریانس σ^2 است.
نویز با نمونه های تبدیل یافته جمع می شود و مستقل از نمونه هاست.

θ : پارامترهای مدل است

$$\mathrm{P}(y_n \mid x_n, \theta) = \mathrm{N}(x_n^T\ \theta\ \ , \sigma^2)$$

$$P(y_n \mid x_n, \theta) = N(x_n^T \theta \ , \sigma^2)$$





ادامه یک مدل احتمالاتی برای LS

به شرط N نمونه درست نمایی (Likelihood) برای داده $Y = (y_1 , y_2 ,..., y_n)$ با داشتن ورودی های X (هر سطر یک داده) و پارامترهای مدل θ به صورت زیر است:

$$P(Y \mid X , \theta) = \prod_{n=1}^{N} P(y_n \mid x_n, \theta) = \prod_{n=1}^{N} N(y_n \mid x_n^T \theta, \sigma^2)$$

ما بایستی این Likelihood را نسبت به پارامترهای مدل θ بیشینه کنیم. یعنی بهترین مدل مدلی است که این درست نمایی را بیشینه کند.

LS , log-likelihood رابطه

Log Likelihood:

$$L_{LL}(\theta) = \log P(y|\, X\,, \theta) = \frac{-1}{2\sigma^2} \sum_{n=1}^{N} (y_n - x_n^T \theta)^2 + \text{constant}$$

LS:

$$L_{MSE}(\theta) = \frac{1}{2N} \sum_{n=1}^{N} (y_n - x_n^T \theta)^2$$

$$\underset{\theta}{\operatorname{argmin}}\, L_{MSE}(\theta) = \underset{\theta}{\operatorname{argmax}}\, L_{LL}(\theta)$$



